INTERVIEW-PREP PACKET 3 OF 5 · ADD-ON · PER-INTERVIEWER · S/4 HANA END TO END · DEFINITIONS AND FORMULAS · V2

Amazon NACF Supply Chain loop — Packet 3 add-on: Rocky Goforth — the S/4 HANA ↔ ISM program end to end

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Thursday, October 30, 2026 · 10:00 am PDT · 55-minute video interview

COMPANION TO PACKET 3 (NINE SEPHORA UDC PRIMARIES). THIS ADD-ON CARRIES THE THREE END-TO-END S/4 STORIES PACKET 3 DOES NOT: THE ONE-UNIT-TO-PUBLISHED-AFS WALK, THE INFRASTRUCTURE VIEW OF THE SAME PROGRAM, AND SHIPPED-NOT-BILLED TO ONE FINANCIAL ON-HAND.

Why this add-on exists. Rocky knows how available-for-sale, Financial On-Hand and the reserve buckets are calculated. A story that says "I unified availability" without the arithmetic will be probed to the definition. So every term below has one exact definition, one system of record, one grain, and every formula is shown symbolic → substituted → computed on a single SKU at the Utah DC across a full day of physical events. **Three new primaries:** S1-R (end to end — one physical unit to the AFS the order system publishes), S1-R-infra (physical truth vs system truth — the infrastructure seat), S3-R (Shipped-Not-Billed and the one Financial On-Hand). Every number is the master canon's value with its basis label; where canon is ambiguous the sources are quoted verbatim and the safe sentence is given instead of a guess.

LP focus — my guess, not confirmed: Dive Deep · Insist on the Highest Standards · Earn Trust · Invent and Simplify. The definitions card and the calculation walk are built for a Dive Deep interviewer; the infrastructure story is built for the title. No interviewer biography is asserted — title-based inference only.

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BACK MATTER

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Canon: amazon/amazon-l6-scm-master-packet.html (S1, S3, S18, §4 ledger) and /tmp/sources/NUMBERS-RECON.md §3 CANON v2. Definitions and formulas trace to the GFiber v8 master (the original ISM chapter: System & data map p36–40, AFS reports p41–45, S3 p62–64) which the Amazon master was transformed from; dates and failure beats to Apple PM III v11 (C1 p22, C2 p24, D1–D2 p28–31, G2 p44–45). Basis labels: record · record-from-docs · user anchor · modeled · illustrative. This file is standalone and does not modify Packet 3.

How to use this add-on — and the 1-page cram sheet NEW

How to use. Read the Definitions card once, slowly. Then read S1-R's calculation walk (Tables W1–W4) with a pen: every row is formula → substitution → value and the Python recompute that built this file asserts every one. S1-R is the story to tell when Rocky asks "walk me through how availability actually got calculated." S1-R-infra is the same program from his seat — scan to WMS to S/4 posting, what breaks when a device or feed is late, the cutover night. S3-R is the Finance leg — Shipped-Not-Billed, de-aggregation to the cent, one Financial On-Hand. Packet 3's U1 (bucket matrix), U3 (merch reserve), U5 (runbook) and U6 (Negative AFS) are the deep dives behind single steps of the walk; this add-on is the arc that connects them.

> **THE ONE DEFINITION AND THE ONE FORMULA — SAY THESE COLD**
>
> **Definition.** "Available-for-sale is the one number a channel may still promise for a SKU at a node. It is not S/4's stock figure. S/4 owns the physical book — unrestricted stock per SKU, plant and storage location, which we called Available. ISM takes that book, applies the lower-of-two rule against the warehouse floor, subtracts everything committed or sold but not yet posted, and publishes AFS per SKU, node and channel. The order system reads it and never computes its own."
>
> **Formula.** "Basis is the lower of S/4 Available and the WMS on-hand; AFS equals Basis minus Picking Reserve minus Ecom Pick minus Ecom Pack minus Shipped-Not-Billed. Merchandise Reserve and Lost-and-Found sit outside Available by construction, so they are never subtracted twice. On the worked SKU: Basis is the lower of 1,000 and 980, so 980; buckets 200 plus 120 plus 60 plus 40 are 420; AFS is 560."
>
> **Financial On-Hand (the honest sentence).** "Financial On-Hand is Finance's number — Available plus Merchandise Reserve plus Lost-and-Found — with Shipped-Not-Billed carried as its own reconciling column so Finance sees what has left the floor but not yet posted; it ties to MB5L at month-end at tolerance zero." FOH expansion: canon says Financial On-Hand; "free on hand" is not in any source — see Verify #1

The three stories in order, the single number each lands

ORDER	STORY	TELL IT WHEN	THE NUMBER TO LAND	BASIS
1	**S1-R** — one unit to published AFS	"How was availability actually calculated?" · Dive Deep · Invent and Simplify	AFS = 980 − 420 = 560; drift 0 at T+1; zero oversell incidents through two go-lives	illustrative walk (canon family) · record
2	**S1-R-infra** — physical truth vs system truth	"What happens when a scan is late or a device fails?" · Insist on the Highest Standards · Are Right, A Lot	WMS feed ≤15 min or publish suppressed; republish p95 <60 s; 343 tasks ≈ 815 h; triggers armed twice, never fired; training +1.5 weeks, date held	record design · record
3	**S3-R** — SNB and one Financial On-Hand	"How did Finance and Supply agree on on-hand?" · Earn Trust · Customer Obsession	FOH truths 2 → 1; Σ per-line COGS ties to the cent ($300.00 = 75 + 50 + 100 + 25 + 50); ≈$0.6M/yr modeled	user anchor · record mechanism · modeled

Three sentences to open with

> "At Sephora two inventory books — the legacy MCS and S/4 HANA — each published their own 'available', and every org defined the word differently. I owned the signed definition and the one engine that publishes it."
>
> "I can walk one unit from the dock to the number the order system reads, bucket by bucket, with the formula at each step — and tell you which report catches it when a step is late."
>
> "The record is zero customer-facing oversell incidents through the January and May go-lives, drift at tolerance zero at T+1, and Finance closing on the same on-hand Operations reads."

The five numbers never to get wrong

1 · **AFS = 980 − (200 + 120 + 60 + 40) = 560** — the canon worked family; flip the floor to 1,020 and Basis is 1,000 → AFS 580, never 600 (illustrative).

2 · **Drift tolerance 0** — ISM S4_Avail − MB52 per SKU × SLoc at the daily watermark; non-zero is a Sev, not a planning variance (record design).

3 · **"January planning go-live, then May execution"** — PO-creation Jan 2025 → combined May 2025 (record, Apple C1 p22). Never "the March go-live."

4 · **$2.5M a year** — the S1 noun: $1M MCS support retired (user anchor) + ≈$1.5M modeled. Always split; never a bare "$2.5M" (it is also S18's monthly slip and S14's order value).

5 · **"Hundreds" of RF devices; training slipped 1.5 weeks; go-live date held** (record, Apple G2). Never "100 of 400"; never a cost multiplier.

Safe-wording rules in force (addendum §D). "About a dozen impacted systems" · "the umbrella program's 836-row log" (it is UDC's, never ISM's) · finished-goods grain only · cutover 343 tasks ≈ 815 task-hours · "$1.5M" and "$2.5M" always with their noun · never a cost multiplier on the RF fallback · FSD dates (3/5, 5/16, 6/5/25) are user anchor — say "signed in the first week of June" if unsure.

Definitions card — built only from the sources NEW

One row per term. "Exact definition" quotes or tightly paraphrases the GFiber v8 ISM chapter (the original definitions the Amazon master S1/S3 were cut from) and the Amazon master; the source column names the page. Where two source passages differ the row says CONFLICT and the Verify list carries the safe sentence. Grain column: N = per node (plant/SLoc or store), C = per channel.

TERM	EXACT DEFINITION (SOURCE WORDING)	SYSTEM OF RECORD	CALCULATED WHEN / WHERE	N / C
Available (S4_Avail)	"S/4 owns the physical book: unrestricted stock per SKU × plant × storage location, which we called Available." MARD-LABST (unrestricted); QI/blocked (INSME/SPEME) are not Available.	SAP S/4HANA MM-IM (MARD / MATDOC; MB52 view)	Posted on every material document; ISM holds a copy: daily MB52 snapshot + MSEG 101/311/601/701/702 lines as real-time events	**N (SKU × WERKS × LGORT); not C**
WMS_OnHand (the floor)	"on-hand by item × location × status … Sum pickable statuses only; damaged/hold excluded"; for OOB the snapshot includes "allocated/picked/packed units (presence, not status), with shipped-not-posted netted via SNB."	BlueYonder WMS	≤15-min feed, key item × whse × as-of; late >15 min ⇒ publish suppressed for that node, last-good AFS held and flagged	**N (DC only; stores have no WMS floor)**
OOB (out-of-balance)	"OOB_in = S4_Avail_in − WMS_OnHand_in (+ = book above floor, − = floor above book)"; "held constant intraday; every closed row carries a cause code (found / confirmed missing → Lost/Found)."	ISM (R6 job, inputs S/4 + WMS)	Daily, both snapshots at the same cutoff — "the one report where a cutoff mismatch manufactures a phantom OOB the size of the day's shipments"	**N**
Basis	"Basis_in = min(S4_Avail_in, WMS_OnHand_in) — a positive OOB reduces supply; a negative OOB is never promised until S/4 books it." Store node: Basis = S4_Avail (no WMS floor).	ISM AFS engine	Recomputed on every event that moves either input	**N**
PickRes (Picking Reserve)	Retail / store-fulfilment commitment: "Σ open STO = PickRes, tolerance 0 (record)"; "PickRes from the STO feed." *Note:* the GFiber mapping row also reads "Commit → Picking Reserve; pick confirm → Ecom Pick; pack confirm → Ecom Pack" — CONFLICT on where a web-order commit lands; R4 (record) is used here. Returns/RTV: "returns post to unrestricted (653/655) and RTV runs as an STO — so they are excluded" from the S1 record.	ISM reserve_ledger (source: S/4 STO EKKO/EKPO type UB; GI via 641)	+ at STO creation; − at 641 goods issue; real-time event, key per STO line	**N (SKU × supplying plant)**
EcomPick	Open e-commerce order units committed/picked, not yet packed: "Σ open orders = EcomPick (+ EcomPack at stores), tolerance 0 (pattern)."	ISM reserve_ledger (source: OMS order-line events)	+ on order commit; moves to EcomPack on pack confirm; cancel reverses and must reference its original; key OMS × OrderLine × line × event_type	**N; published AFS is N × C**
EcomPack	Packed, not yet ship-confirmed e-commerce units. *Note:* R3 says "a Pack balance at a DC must be 0 by definition" while the canon walk carries Ecom Pack 60 at the DC — CONFLICT; the walk follows the canon figures and treats Pack as the packed-not-shipped window.	ISM reserve_ledger (source: OMS pack confirm)	+ on pack confirm; − on ship confirm (moves to SNB)	**N**
SNB (Shipped-Not-Billed)	"SNB_in(t) = Σ sold-or-shipped-not-posted units − Σ matched-posting units. Increment on POS flash sale, OMS omni pending sale, WMS ship confirm; decrement only on a matched S/4 aggregate posting." "SNB ≥ 0 always (a negative attempt is rejected and alerted)."	ISM snb_ledger; S/4 SD/FI owns the posting (Aggregation#, qty, COGS)	+ real-time (Kafka POS line; ship confirm); − on match by Aggregation# ↔ header ↔ lines; nightly for stores, intraday batches at DCs	**N (store or DC where the unit left)**

TERM	EXACT DEFINITION (SOURCE WORDING)	SYSTEM OF RECORD	CALCULATED WHEN / WHERE	N / C
MerchRes (Merchandise Reserve)	"QI/blocked/on-hold → Merch Reserve"; "Merchandise Reserve and Lost/Found are outside S4_Avail by construction, so they never appear in the subtraction — no double-subtract path exists." Visible in FOH and screens, never promisable.	S/4 (stock type) + FRAME holds; ISM displays	Nightly + HUD rollover 00:00 local; Σ reserves ≤ Basis asserted	**N (store: no MerchRes)**
LostFound	"PI differences (701/702) → Lost/Found until cleared."	S/4 (701/702 documents); ISM bucket	On physical-inventory posting; cleared by IC with cause code	**N**
AFS (Available-for-Sale)	"AFS_in = Basis_in − PickRes_in − EcomPick_in − EcomPack_in − SNB_in, where every subtrahend is a committed-or-sold-not-posted bucket." "If AFS < 0: publish 0, write row" (R7). "Sanity: AFS ≤ Basis ≤ both books."	ISM AFS engine — the only publisher; "OMS / dotcom / stores read AFS, never compute"	Event → republish p95 < 60 s; key MATNR × node × channel × version; publish gate: Active Assortment flag per channel	**N × C (published per SKU × node × channel)**
FOH (Financial On-Hand)	S1 (DC): "FOH_n = Avail_n + MerchRes_n + LostFound_n (physical truth for Finance; AFS is the promising truth for channels)"; R11: "(store: no MerchRes)". S3 walkthrough: "available plus merchandise reserve plus lost-and-found plus SNB." CONFLICT on the SNB term — see Verify #1/#2. No source expands FOH as "free on hand."	Published by ISM; S/4 owns value (MBEW-VERPR / WAC); ties to MB5L	Daily after the match cycle; month-end tie to MB5L, "tolerance 0 units; a value difference is a WAC ownership question (S/4 owns it, ISM only displays it)"	**N (SKU × node × day); not C**
Drift (T+1 reconciliation)	"Drift_in = ISM.S4_Avail_in − MB52_in per SKU × SLoc at the daily snapshot; Drift ≠ 0 is a Sev, not a planning variance." Test B identity: "ISM_S4_Avail = (AFS − P) + PickRes + EcomPick + EcomPack + SNB + max(OOB, 0)."	ISM R2 job vs S/4 MB52 / MB5B re-cut by posting time	Daily at a declared watermark; both sides read at the same watermark; tolerance 0	**N**
Negative AFS / AFS Correction	"Negative AFS is a report, not a value channels read"; "clean at T+1 was a go-live exit bar." Every AFS-changing act has "a screen, a role and an audit row."	ISM screens; owners: Negative AFS → Inventory Control + Internal Audit + Finance; AFS Correction → ISM app team	At publish; cleared by cause class (cardinality / latency / override)	**N × C**
Aggregate posting / de-aggregation	"Aggregate posting P with total qty q_P and total COGS C_P; line i with q_i: COGS_i = C_P × q_i / q_P, with Σq_i = q_P and ΣCOGS_i = C_P (allocation is exact; last line absorbs rounding so the sum ties to the cent)." Match rule: "park unless Σ q_i = q_P; >24 h flag, >48 h Sev."	S/4 SD/FI posts; ISM matcher de-aggregates; EDW receives per-line COGS with Aggregation# lineage	Each match cycle; unmatched-aging report (R8) owned by Inventory Accounting	**N × sales-event line**
Publish switch (cutover)	"no SKU may be published from two locations at any instant — zero on the old publish, then 100% on the new" (R10). Rollback: "sync < 99% for four consecutive hours or order-flow p95 > 2× budget for one hour, consumers revert to the prior publish."	ISM publisher; runbook step 4 zero-then-publish	Cutover night, inside a transfer freeze; "armed for both go-lives, never fired"	**N × C**

Sources for every row: GFiber_MASTER_Packet_v8 p36–40 (system map, data-flow table, Math, Data mapping) and p41–45 (R1–R11, "what typically goes wrong"); p62–64 (S3 Math and walkthrough); Amazon master S1/S3 (cb91 p20–21 carry the same formulas verbatim). Movement types 101/311/601/641/653/701/702 and tables MARD/MSEG/MATDOC/MB52/MB5B/MB5L/MBEW are standard SAP named in those pages.

Router addendum — Leadership Principles → these three stories (real AIQB wordings) NEW

Supplements Packet 3's Question → Story Router. Use this grid when the question is about S/4, ISM, availability, Finance tie-out or cutover infrastructure. Question wordings are quoted from the Amazon Interview Question Bank and cited "AIQB: LP Q#".

LP	AIQB QUESTION (VERBATIM)	PRIMARY · ANGLE	BACKUP	15-SECOND OPENER	NUMBER TO LAND
Dive Deep	*"Tell me about a time when you were trying to understand a complex problem on your team and you had to dig into the details to figure it out." (AIQB: Dive Deep Q1)*	S1-R · calculation walk	Packet 3 U6	"Two books each said 'available' and meant different things. I walked one SKU through a day, bucket by bucket, until every unit sat in exactly one place."	**980 − 420 = 560**
Invent and Simplify	*"Give me an example of a complex problem you solved with a simple solution. What made the problem complex?" (AIQB: Invent and Simplify Q1)*	S1-R · design angle	S3-R	"The simple solution was one identity: Basis minus four buckets, one publisher, consumers never compute. The complexity was five feeds at five cadences."	**1 published AFS per SKU × node × channel**
Insist on the Highest Standards	*"Describe a time when you refused to compromise your standards around quality/customer service, etc." (AIQB: Insist on the Highest Standards Q1)*	S1-R-infra · tolerance 0, publish suppression	Packet 3 S18	"Drift tolerance was zero, and when the floor feed went stale we held the last good number and flagged the node rather than compute on a lie."	**≤15 min feed · drift 0 · negatives clean at T+1**
Are Right, A Lot	*"We don't always make the right decision all the time. Tell me about a time when you made a bad decision." (AIQB: Are Right, A Lot Q4)*	S1-R-infra · RF-device script (Apple G2)	master S4	"I scored a device task by effort, not impact; the script failed on new hardware and training slipped a week and a half — the date held."	**+1.5 weeks; date held**
Bias for Action	*"Tell me about a time when you had to gather information and respond immediately to a situation." (AIQB: Bias for Action Q4)*	S1-R-infra · stale feed / negative at cutover	Packet 3 U5	"On the cutover wall a node's floor feed aged past fifteen minutes; the engine had already suppressed publish — we re-read, not re-promised."	**last-good AFS held; 60 s republish**
Earn Trust	*"Tell me about a time when your team's goals were out of alignment with another team you relied on in order to meet your goal." (AIQB: Earn Trust Q5)*	S3-R · Finance keeps its own reconciliation	S1-R	"Finance said 'we keep our own reconciliation.' I showed them the same on-hand arriving from the physical side first, and gave them the aging report to own."	**FOH truths 2 → 1**
Customer Obsession	*"Tell me about a time when you had to balance the needs of the customer with the needs of the business." (AIQB: Customer Obsession Q8)*	S3-R · decrement at sale, not at posting	S1-R	"A store sale S/4 would not see until the nightly aggregate was a unit dotcom could still promise. I moved the decrement to the moment of sale."	**phantom window → 0; late line = −2, never +2**
Ownership	*"Describe a time when you had to transition a project you owned to a new owner." (AIQB: Ownership Q5)*	S3-R · Finance owns the aging report	Packet 3 U6	"The product was not the matcher; it was Inventory Accounting running the unmatched-aging report without me."	**>24 h flag · >48 h Sev · match ≥99.5% design target**
Deliver Results	*"Tell me about a time when you had significant, unanticipated obstacles to overcome in achieving a key goal." (AIQB: Deliver Results Q2)*	S1-R-infra · cutover night	Packet 3 U5	"343 tasks, about 815 hours, one irreversible instant — zero then publish — with rollback triggers armed twice and never fired."	**343 / ≈815 h / never fired**
Have Backbone; Disagree and Commit	*"Describe a time when you took an unpopular stance in a meeting with peers and your leader." (AIQB: Have Backbone Q2)*	S1-R · lower-of-two (surplus never promised)	Packet 3 U3	"Merchandising wanted the 20-unit floor surplus promised. I said no: a negative OOB is never promised until S/4 books it — AFS 580, not 600."	**580, not 600**

Non-LP prompts: "explain a formula you own" → S1-R Table W1; "a time Finance disagreed with ops" → S3-R; "a go-live gone wrong" → S1-R-infra (device script); "how do you know two systems agree" → S1-R Table W4 (Test A / Test B).

The stories — end to end, definitions first, canon numbers

Three primaries on the Sephora S/4 HANA ↔ ISM program. Each: 60-second pitch → Situation/Task → Action in five gated phases → Results → L6 evidence → five facets and probing → the calculation walk and four-table math → behavioral pack, angle cards, trap drills, 20-second version, NACF bridge and the safe-say box. The worked SKU is illustrative and uses the canon 1,000 / 980 / 560 family so it matches Packet 3's U1, U3 and U6.

S1-R — End to end: from one physical unit at the Utah DC to the AFS the order system publishes — two books, one signed definition, one engine, one identity that must hold at T+1 with tolerance zero NEW

LPs this story serves here: Dive Deep (primary — the calculation walk) · Invent and Simplify (one identity, one publisher) · Have Backbone; Disagree and Commit (the surplus is never promised) · Earn Trust (three orgs signed one definition)
Process family: Plan / Deliver — inventory availability definition and control: bucket design, system of record per bucket, publish contract, reconciliation

Honesty label. Record: two books (MCS and S/4) with "available" defined differently by different orgs; recurring oversell → zero customer-facing incidents through both go-lives; the FSD-signed policy (lower-of-two basis, AFS/FOH equations, real-time decrement contract, daily ISM-vs-MB52 reconciliation at tolerance zero); PO-creation go-live January 2025 → combined go-live May 2025; E2E chain Inbound 11/22 · Dotcom 12/20 · Retail 1/30; 60+ stakeholders; about a dozen impacted systems. User anchor (GFiber v8 only — confirm): FSD v1.1 3/5/25 → v1.2 5/16/25 → signed 6/5/25; three signing orgs; $1M MCS support retired. Modeled: ≈$1.5M (oversell $0.9M + reconciliation $0.33M + change premium $0.3M) on an illustrative 30M-order base. Illustrative: every unit count in the walk (SKU 12345 × node 801, the 240-unit receipt, the 700-unit damage, the −30). I wrote the definitions, the identity, the keys, the tolerance and the owner map; the S/4 and ISM engineers built the extracts, the engine and the screens; Inventory Control, Inventory Accounting and the ISM app team run the controls.

LEADERSHIP PRINCIPLES CARRIED — ONE-LINE PROOF EACH

Dive Deep. I can walk one unit from the 101 receipt to the number the order system reads, name the bucket it sits in at every minute, and name the report that catches it when a step is late.

Invent and Simplify. Five feeds at five cadences collapsed into one identity — Basis minus four buckets — with one publisher and consumers that never compute.

Have Backbone; Disagree and Commit. I refused to promise the floor surplus: a negative OOB is never promised until S/4 books it — 580, not 600.

Earn Trust. Three orgs that defined "available" three ways signed one table; Finance reads the same on-hand Operations reads.

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. Sephora ran two inventory books — the legacy MCS and S/4 HANA — and every org meant something different by "available": Finance meant the posted book, dotcom meant what it could promise, the DC meant what it could pick. The result was recurring oversell — units promised that the floor could not back — and a negative available-to-sell that surfaced as an order that could not ship. **Task.** Own one signed definition of available, one engine that publishes it, and the controls that prove the two books agree every day. **Action.** I defined the buckets and the system of record for each: S/4 owns the physical book — unrestricted stock per SKU, plant and storage location; the WMS owns the floor count; the OMS owns what is committed, picked and packed; the POS owns what a store sold that S/4 will not see until the nightly aggregate. ISM takes the lower of book and floor as the basis, subtracts Picking Reserve, Ecom Pick, Ecom Pack and Shipped-Not-Billed, and publishes one AFS per SKU, node and channel; the order system reads it and never computes its own. Reconciliation is daily, ISM's copy of available against MB52, tolerance zero — a non-zero drift is a Sev, not a planning variance. Three orgs signed it in the FSD; January was the planning go-live, May the execution go-live. **Result.** Primary metric: customer-facing oversell incidents from recurring to zero through both go-lives, negatives clean at T+1 at cutover, drift at zero; about $2.5M a year — $1M of MCS support retired, my program number, and roughly $1.5M modeled, inputs on request.

SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

S — Two books, one word. MCS and S/4 each published an "available"; Finance, Merchandising, dotcom, the stores and the DC each defined it differently; the stores' sales reached S/4 only as a nightly aggregate, the OMS committed units S/4 could not see, and the WMS floor disagreed with the book by whatever had moved since the last snapshot. Oversell was recurring — no count in the record — and reconciliation was argument.

T — Write the one definition of available that every org would sign; assign one system of record per bucket; make one engine publish it per SKU × node × channel; and prove daily, at tolerance zero, that the engine's copy of the book equals the book.

Baselines on the table: 2 books · "available" defined per org · oversell recurring (record, no count) · reconciliation by hand (≈4 FTE, benchmark) · 60+ stakeholders, about a dozen impacted systems (record) · a 30M-order illustrative base for the model.

ACTION — THE PROCESS / OPERATING-MODEL REDESIGN, IN GATED PHASES

- **Discovery (3/29 → 4/30)** — bucket inventory: every place a unit could sit, who owned it, which document created it; the five feeds and their cadences (S/4 documents, WMS ≤15 min, OMS real-time, POS real-time, S/4 aggregate nightly). *Exit bar:* every bucket has exactly one owning system; open questions dated.
- **Plan & Design (5/15 → 8/14; ISM FSD v1.1 3/5/25 → v1.2 5/16/25 → signed 6/5/25 — user anchor)** — the lower-of-two basis; AFS and FOH equations; the real-time decrement contract (event → one bucket delta; idempotency key per event; a reversal must reference its original); the daily drift check at tolerance zero; the control-owner map. *Exit bar:* the sellable table signed by the business; every AFS-changing act has a screen, a role and an audit row.
- **Build & Test (Aug → Jan/Apr)** — the node × channel × bucket-transition matrix as the regression asset (Packet 3 U1); the replay test (the day replayed into a shadow ledger equals production); E2E chain Inbound 11/22 · Dotcom 12/20 · Retail 1/30. *Exit bar:* Blocker + Critical open = 0; three E2E dates green.
- **Cutover & Go-Live (January planning go-live; May execution go-live)** — R10 cutover balance sheet (MCS closing · S/4 opening MB52 · ISM seed, Σ per bucket equal); clear negatives first; zero-then-publish so no SKU is ever published from two locations; OMS re-pointed to read ISM AFS; rollback triggers armed. *Exit bar:* negatives clean at T+1; no SKU with two published AFS values; drift 0 on the first morning.
- **Hypercare / Close (Jun → Jul)** — I read the drift report every morning until the control owners read it without me; MCS retired (no live arrow). *Exit bar:* drift = 0 SKUs on consecutive days; owners running R2, R6, R7, R8 without me.

Influence and ownership. Mine: the definitions, the identity, the keys, the tolerance, the owner map, the go/no-go recommendation. The S/4 and ISM engineers built the extracts, the engine and the screens; Inventory Control owns drift and OOB; Inventory Accounting owns the settlement match; the ISM app team owns replay, quarantine and AFS Correction.

> **Designed the process; others own it now.** The daily drift report, the Negative AFS and AFS Correction screens, and the unmatched-aging report are run by their owners inside the monthly release train. **Trade-off put to leadership:** an asymmetric basis — surplus never promised, so visible undersell and more investigations — against oversell as a customer defect. Leadership accepted the undersell.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY

- **$** — about $2.5M a year: $1M MCS support retired (user anchor) + ≈$1.5M modeled (oversell 30,000,000 × 0.10% × $30 = $900,000; reconciliation 3 × $110,000 = $330,000; change premium 40 × 60 h × $125 = $300,000). Never a bare "$2.5M".
- **#** — 2 books → 1 published AFS per SKU × node × channel; 4 subtrahend buckets; 11 reports (R1–R11) each with a grain, a watermark, a tolerance and an owner; three signing orgs (user anchor).
- **%** — drift tolerance 0%; design: feed completeness ≥99.9%; the model's 0.10% oversell rate is a benchmark, not a measurement.
- **Time** — January planning go-live → May execution go-live (record); event → republish p95 < 60 s; WMS feed ≤ 15 min; negatives clean at T+1.
- **Customer** — customer-facing oversell incidents recurring → zero through both go-lives (record).
- **Quality** — every booked unit sits in exactly one bucket: ISM_S4_Avail = (AFS − P) + PickRes + EcomPick + EcomPack + SNB + max(OOB, 0), asserted daily (record design).

L6 LEVEL EVIDENCE — every line is this story's record or labeled model, no new canon numbers

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	"Available" meant three things to three orgs; I wrote the one table and got it signed.
Scope & influence	60+ stakeholders across about a dozen impacted systems; none reported to me; three orgs signed.
Execution trade-off + mechanism others own	Asymmetric basis (undersell accepted) against oversell; the mechanism others own: R1–R11 with named owners.
Impact	Oversell recurring → zero through two go-lives; MCS retired; Finance and Operations on one number.
Process improvement	Reconciliation moved from argument to a daily identity at tolerance zero; a new node or channel is a matrix row, not a project.
Complexity & consensus	Five feeds at five cadences; one identity; one publisher; signed FSD.
Root cause at scale	Every break is attributable: timing (posting time > watermark), duplicate (transition log), true drift (IC with cause code).

PROBING — FIVE FACETS, THEN THE HARDEST FOLLOW-UPS NEW

THE FIVE FACETS — WHERE EACH ONE LIVES IN THIS STORY

1 · Data analysis. Grain: SKU × node (plant/SLoc) × channel × as-of timestamp. Data: MARD-LABST unrestricted per SKU × WERKS × LGORT (daily MB52 snapshot); MSEG lines 101/311/601/641/701/702 as events keyed MBLNR × ZEILE; WMS on-hand by item × location × status (≤15 min); OMS order-line events keyed OMS × OrderLine × line × event_type; POS RetailSaleTransactions keyed Txn × line; S/4 aggregate postings keyed Aggregation#. Joins: MATNR × WERKS × LGORT ↔ ISM node via the node map (801 inventory / 800 selling location is why the map is a table, not a rule); Aggregation# ↔ POS header ↔ lines. Revealed: the books legitimately disagree at every instant because of cadence, so the disagreement must be explained by an identity, not argued.

2 · Solution design. Options: (A) make S/4 the promising truth — rejected, it cannot see a 09:14 web order or a 15:30 parcel before the aggregate posts; (B) let each consumer compute its own available from the feeds — rejected, that is the two-dialect problem again; (C) one engine beside S/4 publishes AFS, consumers never compute, and S/4 keeps the book — chosen. Inside C: symmetric vs asymmetric basis — asymmetric chosen (surplus never promised).

3 · Math & data logic. OOB = S4_Avail − WMS_OnHand; Basis = min(S4_Avail, WMS_OnHand); AFS = Basis − PickRes − EcomPick − EcomPack − SNB; FOH = Avail + MerchRes + LostFound; Drift = ISM.S4_Avail − MB52 = 0; identity ISM_S4_Avail = (AFS − P) + PickRes + EcomPick + EcomPack + SNB + max(OOB, 0). Worked below on one SKU through twelve events.

4 · Implementation. Five gated phases above; cutover: clear negatives → transfer AFS to Uncommitted → scan-and-reserve → zero-then-publish; rollback triggers per interface; hypercare exit at drift 0 on consecutive days. Risks: vocabulary drift, feed latency, override abuse.

5 · Probe questions. Below: is FOH before or after reserves; why isn't S/4 the truth; what breaks at the watermark; who owns the publish; and the trap drills.

FACET 5 — PROBE QUESTIONS (CHALLENGE Q&A TUNED TO THIS INTERVIEWER)

Infra — where is the single source of truth for "available," and why isn't it S/4?

Physical stock truth is S/4, full stop. Promisable truth is ISM's published AFS, because S/4 posts documents and settles store sales as a nightly aggregate while promising needs every click and scan. The two are reconciled, not merged: ISM's copy of Available must equal MB52 daily at tolerance zero, and Financial On-Hand must tie to MB5L at month-end.

Infra — is Financial On-Hand before or after reserves?

It includes them. FOH = Available + Merchandise Reserve + Lost-and-Found — it is Finance's physical number, so a held unit is still on hand. AFS is the number after reserves and after every committed bucket. Reserve sits outside Available by construction, so it is never subtracted from AFS twice. Shipped-Not-Billed is carried as its own column so Finance can see what has left the floor but not yet posted.

see Verify #1–2 on the SNB term

Infra — what is the join key between S/4 and ISM, and where does it drift?

MATNR × WERKS × LGORT maps one-to-one to an ISM node through the node map; the sold-not-posted bucket joins on Aggregation# back to POS lines. It drifts in two places: an unmapped movement type — a new BWART nobody told ISM about, quarantined, never guessed — and a late goods receipt that crosses the snapshot watermark, which is why the drift report attributes timing rows before anyone calls a Sev.

Infra — who owns the publish?

ISM's AFS engine is the only publisher — per SKU × node × channel, key MATNR × node × channel × version, event to republish p95 under 60 seconds. The OMS, dotcom and the stores read it and never compute. At cutover no SKU could be published from two locations at any instant: zero on the old publish, then 100% on the new.

Infra — why not just trust MARD?

On S/4 MARD is trustworthy — it is a view over MATDOC, not a drifting aggregate — but it answers "what is booked," not "what may be promised." It cannot see a web order placed at 09:14 or a parcel that left at 15:30 before the aggregate posts. So I used it as the anchor of the identity and built the reports to prove every other term against it.

BUSINESS IMPACT MODEL — KPI DRIVERS, BASELINE → AFTER

DRIVER	BASELINE	AFTER	$ / EFFECT	BASIS
Published "available" values per SKU	2 books, per-org dialects	1 AFS per SKU × node × channel	—	record
Oversell incidents (customer-facing)	recurring (no count)	0 through both go-lives	≈$0.9M/yr exposure (30M × 0.10% × $30)	record / modeled
MCS support	$1M/yr	retired	$1.0M/yr	user anchor
Reconciliation	argument; ≈4 FTE by hand	daily identity, tolerance 0; ≈1 FTE reading it	≈$0.33M/yr	modeled
Change premium (two books to keep in step)	40 changes × 60 h × $125	one engine	≈$0.3M/yr	modeled
Total	—	—	≈$2.5M/yr = $1.0M anchor + ≈$1.53M modeled	split always

THE CALCULATION WALK — ONE SKU, ONE NODE, ONE DAY: FORMULA → SUBSTITUTION → VALUE AT EVERY STEP NEW

Setup (illustrative, canon family). SKU 12345 at the Utah DC, ISM node 801 (dotcom inventory location; selling location 800), the day after the 06:00 MB52 snapshot. Opening: S/4 Available 1,000; WMS on-hand 980 — the floor count with shipped-not-posted netted back per R6, so 980 = 940 units physically pickable + 40 units shipped-not-posted; Picking Reserve 200 (open STOs to stores); Ecom Pick 120; Ecom Pack 60; Shipped-Not-Billed 40 (omni pending-sale lines that settle on the nightly aggregate); Merchandise Reserve 150 and Lost/Found 0 outside Available. FRAME channel reserve is 0 in this walk, so the dotcom channel's published AFS equals the node AFS (the per-channel priority walk is S2 / Packet 3 U1). Promisable in-transit P = 0 at this node. Every row below is asserted by the recompute script.

Table W1 — the day's event ledger: which physical event moves which bucket, and what AFS does.

#	TIME · PHYSICAL EVENT → SYSTEM EVENT	S4_AVAIL	WMS FLOOR (NETTED)	OOB	BASIS	PICKRES / ECOMPICK / ECOMPACK / SNB	Σ BUCKETS	AFS
E0	06:00 MB52 snapshot; ISM copy refreshed (R1)	1,000	980	+20	min(1,000, 980) = 980	200 / 120 / 60 / 40	420	980 − 420 = 560
E1	07:30 trailer unloaded; PO goods receipt 101 for 240 EA posts in S/4; WMS holds them in receiving-staging status (not pickable)	1,000 + 240 = 1,240	980 (staging excluded)	1,240 − 980 = +260	min(1,240, 980) = 980	200 / 120 / 60 / 40	420	980 − 420 = 560 (dock units not promisable)
E2	09:10 putaway complete; WMS status → pickable; feed lands inside 15 min	1,240	980 + 240 = 1,220	1,240 − 1,220 = +20	1,220	200 / 120 / 60 / 40	420	1,220 − 420 = 800 (republished < 60 s)
E3	09:14 dotcom order committed, 30 EA (OMS commit event)	1,240	1,220	+20	1,220	200 / 120 + 30 = 150 / 60 / 40	450	1,220 − 450 = 770
E4	10:02 STO created for store replenishment, 100 EA (retail channel)	1,240	1,220	+20	1,220	200 + 100 = 300 / 150 / 60 / 40	550	1,220 − 550 = 670
E5	11:20 RF pick confirm and pack confirm, 50 EA of the open ecom lines	1,240	1,220	+20	1,220	300 / 150 − 50 = 100 / 60 + 50 = 110 / 40	550	670 — unchanged: bucket-to-bucket
E6	13:05 WMS ship confirm, 90 EA leave the dock (parcel) → SNB += 90	1,240	1,090 physical + 130 SNB = 1,220	+20	1,220	300 / 100 / 110 − 90 = 20 / 40 + 90 = 130	550	670 — unchanged: already promised
E7	13:20 S/4 delivery goods issue 601 posts for the 90 (intraday); matched to the ship confirm → SNB −= 90	1,240 − 90 = 1,150	1,090 + 40 = 1,130	1,150 − 1,130 = +20	1,130	300 / 100 / 20 / 130 − 90 = 40	460	1,130 − 460 = 670 — unchanged
E8	14:00 STO goods issue 641, 100 EA on the truck to the store (in-transit shows on MB5T)	1,150 − 100 = 1,050	990 + 40 = 1,030	1,050 − 1,030 = +20	1,030	300 − 100 = 200 / 100 / 20 / 40	360	1,030 − 360 = 670 — unchanged
E9	16:10 rack leak: WMS moves 700 EA to damaged/hold status; S/4 transfer posting awaits IC approval	1,050	290 + 40 = 330	1,050 − 330 = +720	min(1,050, 330) = 330	200 / 100 / 20 / 40	360	330 − 360 = −30 → publish 0; Negative AFS row (R7)
E10	16:40 IC approves; S/4 posts unrestricted → blocked 700 EA (→ MerchRes 150 + 700 = 850)	1,050 − 700 = 350	330	350 − 330 = +20	330	200 / 100 / 20 / 40	360	−30 — the over-commit is real: 30 promised EA cannot ship from 801
E11	16:55 AFS Correction (role + audit row, cause class "latency/physical"): OMS re-sources 30 EA of open ecom lines to node 901	350	330	+20	330	200 / 100 − 30 = 70 / 20 / 40	330	330 − 330 = 0 — negative clean before T+1
E12	17:00 watermark: T+1 reconciliation (R2) — see Table W4	350 = MB52 350	330	+20	330	200 / 70 / 20 / 40	330	0; Drift = 350 − 350 = 0

Reading the ledger: a receipt moves S/4 first and the floor second (E1 → E2), so AFS waits for putaway — time-to-available is the dock-to-stock window. Commit and STO creation move AFS (E3, E4); pick, pack, ship and goods issue do not (E5–E8) — they move units between buckets or out of both books at once. A physical loss moves AFS instantly through the floor (E9), and the book catches up on a document (E10). The published number is never negative; the negative is a row with an owner (E9–E11).

Table W2 — the four formulas, symbolic → substituted → computed, at three instants.

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	OOB at E0	OOB = S4_Avail − WMS_OnHand = 1,000 − 980	record formula	+20 (book above floor)
2	Basis at E0	Basis = min(S4_Avail, WMS_OnHand) = min(1,000, 980)	record formula	980
3	AFS at E0	AFS = Basis − PickRes − EcomPick − EcomPack − SNB = 980 − 200 − 120 − 60 − 40	record formula; canon walk	560
4	Flip the floor (asymmetry)	WMS 1,020 → OOB = 1,000 − 1,020 = −20 → Basis = min(1,000, 1,020) = 1,000 → AFS = 1,000 − 420	canon walk	580, not 600 — surplus unpromised
5	AFS at E2 (after putaway)	Basis = min(1,240, 1,220) = 1,220; AFS = 1,220 − (200 + 120 + 60 + 40)	illustrative	800

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
6	Real-time decrement E2 → E4	AFS(t) = AFS(t_0) − Σ Δ_e = 800 − 30 (commit) − 100 (STO)	record contract	670
7	Bucket-to-bucket conservation E5–E8	Σ buckets − (Basis change) : 550 → 550 → 460 (−90 with Basis −90) → 360 (−100 with Basis −100); AFS 670 each time	illustrative	670 unchanged
8	FOH at E0 (DC)	FOH = Avail + MerchRes + LostFound = 1,000 + 150 + 0	record formula (S1); SNB term see Verify #2	1,150
9	FOH at E12	350 + 850 + 0	record formula	1,200
10	Physical cross-check at E12	present = 290 pickable + 700 damaged + 150 reserve = 1,140; + 40 shipped-not-posted = 1,180; FOH − 1,180 = 20 = the open OOB row	illustrative; OOB row to IC with cause code (R6)	20 = OOB

Table W3 — what goes wrong, one variant per control, with the number.

CONTROL	VARIANT (WHAT GOES WRONG)	FORMULA → SUBSTITUTION	CAUGHT BY · OWNER	EFFECT ON AFS
R9 idempotency	The 13:05 ship confirm for 90 EA is replayed at 13:07	same key OMS × OrderLine × line × ship → no-op; AFS stays 670. If applied: EcomPack 20 − 90 = −70 (rejected: bucket < 0) / SNB 130 + 90 = 220 → AFS 1,220 − 640 = 580	R9 replay audit · ISM app team	0 (would have been −90, safe direction)
WMS feed SLA	At E2 the floor feed ages past 15 min	publish suppressed for node 801; last-good AFS 560 held and flagged; the 240 dock units stay unpromised; undersell bound = 240 × $4.75 = $1,140 if all 240 would have sold in the window	feed-latency monitor · WMS team	held at 560 (undersell, never oversell)
R2 Test A (timing)	Extract pulled 17:05; a 101 for 24 EA posted 17:03	S/4 374 − ISM 350 = +24 → MATDOC CPUTM 17:03 > watermark 17:00 → timing; MB5B re-cut at 17:00 → 350 − 350 = 0	R2 drift · ISM engineers fix the extract (cut by posting time)	0 after re-cut; no Sev
R6 cutoff	WMS snapshot at 17:00, S/4 snapshot at 13:00	phantom OOB = day's goods issues = 90 + 100 = 190; Basis understated by 190 → AFS 0 − 190 → published 0 and a false negative row	R6 same-cutoff rule · Inventory Control	−190 phantom (safe direction, but a false Sev)
R7 Negative AFS	E9: 700 EA to damaged while 360 committed	AFS = 330 − 360 = −30 → publish 0, write row; exposure 30 × $30 = $900	R7 · Inventory Control + Internal Audit + Finance; fix via AFS Correction	−30 → 0 at E11
R4 tie-out	OMS shows 31 open ecom lines for 801 after E11, ISM EcomPick 70 for 70 units of 31 lines; STO side Σ open STO = 200	Σ open STO 200 − PickRes 200 = 0 ✓; Σ open orders 70 − EcomPick 70 = 0 ✓ — a break here is a lost or duplicated event, never a units problem	R4 · ISM engineers first	0
R8 unmatched	Nightly aggregate for the 40 omni lines arrives as 38 EA	Σq_i 40 ≠ q_P 38 → park; SNB stays 40; S4_Avail drops 38 → AFS under-stated by 2 until the lines land; >24 h flag, >48 h Sev	R8 unmatched-aging · Inventory Accounting	−2 (safe direction)
R10 publish switch	Cutover: SKU 12345 found published from both 800 and 900	rule: published locations per SKU = 1 at every instant; zero on old → 100% on new; Σ AFS conserved across the switch	R10 balance sheet · war room	0 double-publish allowed

Table W4 — T+1 reconciliation: the two tests on the worked row (watermark 17:00).

STEP	TEST	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Test A — external: book vs copy	MB52_Avail − ISM.S4_Avail = 350 − 350	record; tolerance 0; ≠ 0 is a Sev until classified	0 ✓
2	Test B — internal identity	ISM_S4_Avail = (AFS − P) + PickRes + EcomPick + EcomPack + SNB + max(OOB, 0) = (0 − 0) + 200 + 70 + 20 + 40 + 20	record identity	350 ✓
3	Test B at E0 (sanity)	(560 − 0) + 200 + 120 + 60 + 40 + 20	canon family	1,000 ✓
4	Why both tests	a duplicate 601 consumed twice: ISM 340 vs MB52 350 → Test A −10, Test B still ✓ (engine did correct arithmetic on a wrong input)	record lesson (GFiber R2 row 3)	Test A catches it
5	Negative AFS clean at T+1	rows with AFS < 0 at 17:00 for node 801 = 0	record exit bar	0 ✓
6	Month-end	Σ_n FOH_n × MBEW-VERPR ties to MB5L; tolerance 0 units; a value gap is a WAC question (S/4 owns)	record mechanism; days-to-tie not in record	ties

THE MATH — FOUR-TABLE DERIVATION (DOLLARS; FORMULA → SUBSTITUTION → RESULT) NEW

Table 1 — before: the problem in numbers.

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Published "available" values per SKU	`MCS + S/4 = 2, each read by different orgs`	record	2
2	Oversell exposure (baseline)	`30,000,000 orders × 0.10% × $30`	illustrative base; benchmark rate; $30 all-in	$900,000/yr
3	Reconciliation labor	`≈4 FTE × $110,000`	benchmark	$440,000/yr
4	MCS support	`$1,000,000`	user anchor	$1.0M/yr
5	Change premium	`40 changes × 60 h × $125`	benchmark	$300,000/yr

Table 2 — the fix: step by step, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	One basis	`Basis = min(1,000, 980)`	record	980
2	One AFS	`980 − (200 + 120 + 60 + 40)`	record	560
3	One publisher	`published AFS values per SKU × node × channel = 1; consumers compute = 0`	record	1
4	One identity, daily	`Drift = 350 − 350 = 0; tolerance 0`	record	0
5	Reconciliation after	`≈1 FTE × $110,000`	benchmark	$110,000/yr

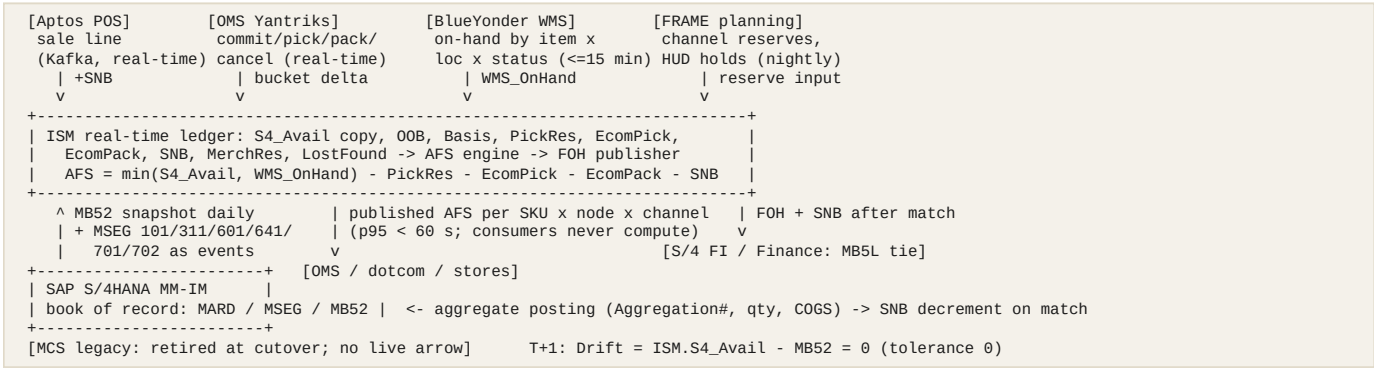
Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Books publishing "available"	2	1 (S/4 keeps the physical book; ISM publishes)	record
2	Customer-facing oversell incidents	recurring	0 through both go-lives	record
3	Drift ISM vs MB52	unmeasured	0, tolerance 0, daily	record design
4	Negative AFS at cutover	present (MCS: e.g., 400 of 1,800 SKUs in one department — issues-log row 716, UDC log)	clean at T+1	record-from-docs / record
5	Reconciliation FTE	≈4	≈1	benchmark

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Oversell avoided	`30,000,000 × 0.001 × $30 = $900,000`	modeled	$0.90M
2	Reconciliation released	`(4 − 1) × $110,000 = $330,000`	modeled	$0.33M
3	Change premium	`40 × 60 × $125 = $300,000`	modeled	$0.30M
4	Modeled subtotal	`900,000 + 330,000 + 300,000 = $1,530,000`	modeled	≈$1.5M
5	MCS support retired	`$1,000,000`	user anchor	$1.0M
6	Headline	`1,000,000 + 1,530,000 = $2,530,000`	anchor + modeled — always split	≈$2.5M/yr
7	Exposure of the worked negative	`30 × $30 = $900`	illustrative	$900

SOLUTION FLOW — THE SYSTEM & DATA MAP



SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Oversell rate (30M orders × $30)	0.05% → $450K	0.10% → $900K	0.25% → $2.25M

INPUT	LOW	BASE	HIGH
Reconciliation FTE released × $110K	2 → $220K	3 → $330K	4 → $440K
Change premium (changes × 60 h × $125)	20 → $150K	40 → $300K	60 → $450K
Modeled total (excl. $1M anchor)	$820K	$1.53M	$3.14M
Dock-to-stock window on the 240 EA (E1 → E2)	40 min	100 min	4 h → publish held longer; undersell only

SPOKEN STATEMENT

"Two books each said 'available' and meant different things, so I wrote the one definition and got it signed. S/4 owns the physical book — unrestricted stock per SKU, plant and storage location. ISM takes the lower of that book and the warehouse floor as the basis, subtracts Picking Reserve, Ecom Pick, Ecom Pack and Shipped-Not-Billed, and publishes one available-for-sale per SKU, node and channel; the order system reads it and never computes. On the worked SKU that is the lower of 1,000 and 980, less 420 committed, 560. A receipt does not move it until putaway; a commit moves it in under a minute; a pick, a pack or a ship moves units between buckets and leaves it alone; a loss on the floor moves it instantly and the book catches up on a document. Every morning ISM's copy of the book must equal MB52 at tolerance zero — a difference is a Sev, not a planning variance — and every booked unit must sit in exactly one bucket. The record is oversell from recurring to zero through both go-lives and negatives clean at T+1; the dollar view is about $2.5M a year — $1M of MCS support retired, and roughly $1.5M modeled."

IF CHALLENGED

"Your oversell number is the biggest modeled piece — how do you know it was a tenth of a percent?"

I don't know it to the decimal. The record says the incidents were recurring and went to zero; the rate is a benchmark I set deliberately low. The first thing I'd measure is the OMS cancel-reason code "item unavailable after order" for the twelve months before January.

"The ship confirm and the goods issue both touch SNB — isn't that double counting?"

No — they are the two halves of one movement. The ship confirm adds 90 to SNB and removes 90 from the floor; the floor is netted by SNB for the OOB comparison, so Basis is unchanged and so is AFS. When the 601 posts, S/4 Available drops 90 and the matched SNB drops 90 — Basis falls 90 and the buckets fall 90 together. AFS is 670 before, during and after. The walk shows it at E6 and E7.

TWIST MAP — LIKELY LP QUESTIONS FROM THIS INTERVIEWER → THE ANGLE TO TAKE

LIKELY QUESTION (THIS INTERVIEWER)	LP	ANGLE — WHAT TO EMPHASIZE	SKIP / NEVER SAY
Walk me through how availability was actually calculated.	Dive Deep	Table W1 from the top: E0 → E12, name the bucket at each step.	Never quote the unit counts as program data.
What was the simplest thing you did on a complex program?	Invent and Simplify	One identity, one publisher, consumers never compute.	Do not claim I built the engine.
Tell me about a time you said no to a stakeholder.	Have Backbone	The surplus is never promised: 580, not 600.	Do not say Merchandising was wrong — they had a real visibility need (U3).
How did you get three orgs to agree?	Earn Trust	A table, not a meeting: term, definition, system of record, grain, owner — signed.	Say "three orgs" only if confident; names are user anchor.

BEHAVIORAL PACK — FOUR SEPARATE 90-SECOND STAR ANSWERS, EACH FROM A DIFFERENT MOMENT OF THE STORY

NEW

DIVE DEEP · AIQB Q1 **Q1. "Tell me about a time when you were trying to understand a complex problem on your team and you had to dig into the details to figure it out. Who did you talk with or where did you have to look to find the most valuable information?"**

The complex problem was that two inventory books disagreed every day and nobody could say which was wrong. I stopped asking people what "available" meant and went to the documents: MARD unrestricted by SKU, plant and storage location on the S/4 side; the WMS on-hand by status; the OMS order-line events; the POS transaction lines; and the nightly aggregate posting that is the only way a store sale reaches S/4. Then I walked one SKU through one day — a receipt at 07:30 that S/4 sees before the floor does, a putaway at 09:10 that makes it promisable, a web commit at 09:14, a store transfer at 10:02, a pick and pack that move units between buckets, a ship confirm and a goods issue that move them out of both books together. That walk showed the books legitimately disagree at every instant because of cadence, so the fix was not a better spreadsheet but an identity: every booked unit sits in exactly one bucket, asserted daily at tolerance zero. The most valuable information came from Inventory Control's morning reconciliation and the ISM transition log — the people and the table that saw every break.

Number to land: Basis min(1,000, 980) = 980; AFS 980 − 420 = 560; drift 0 · **Pivot to NACF:** At an FC the same walk is receive scan → stow → available: I would ask which system is the book, which is the floor, and what the identity between them is. · **Do NOT say:** that the unit counts are program data; that I built the extracts.

INVENT AND SIMPLIFY · AIQB Q1 **Q2. "Give me an example of a complex problem you solved with a simple solution. What made the problem complex? How do you know your solution addressed the problem?"**

The complexity was five feeds at five cadences — S/4 documents, a 15-minute warehouse feed, real-time order events, real-time store sales, and a nightly aggregate posting — and about a dozen impacted systems each tempted to compute its own available. The simple solution was one identity and one publisher. Basis is the lower of the book and the floor; available-for-sale is Basis minus four committed buckets; ISM publishes it per SKU, node and channel; everyone else reads it and never computes. Merchandise reserve and lost-and-found sit outside the book by construction so no double subtraction is possible. I know it addressed the problem because the identity is checkable: ISM's copy of the book equals MB52 every morning at tolerance zero, and the book equals AFS plus the buckets plus the positive out-of-balance. The record is oversell from recurring to zero and negatives clean at T+1 at cutover.

Number to land: 1 published AFS per SKU × node × channel; identity 350 = 0 + 200 + 70 + 20 + 40 + 20 · **Pivot to NACF:** At NACF I would want one published available per ASIN × FC that placement and transfer read and never recompute. · **Do NOT say:** "simple" meant "easy" — it took a signed FSD and a 343-task cutover.

HAVE BACKBONE; DISAGREE AND COMMIT · AIQB Q2 **Q3. "Describe a time when you took an unpopular stance in a meeting with peers and your leader. What was it? Why did you feel strongly about it?"**

The stance was that a surplus on the warehouse floor is never promised until S/4 books it — and the room wanted the units. If the floor shows 1,020 and the book shows 1,000, the symmetric rule says promise 1,020 less commitments; I set Basis to the lower of the two, so AFS is 580, not 600, and twenty sellable units sit unpromised. The argument against me was lost sales. My argument was asymmetry: an oversell is a customer we have already told yes and cannot ship — about thirty dollars on benchmark inputs; an undersell is a planning variance Inventory Control chases through the out-of-balance report. With no measured phantom rate at design time I set the probability that a surplus is phantom to one and let the Negative AFS and correction screens chase the undersell. Leadership accepted the undersell. I committed to the cost — more investigations — and the record is zero oversell through both go-lives.

Number to land: 580, not 600; OOB −20 never promised · **Pivot to NACF:** Same stance on dock-received versus system-available: I would not promise a unit that is scanned but not stowed. · **Do NOT say:** that the lost sales were zero; that Merchandising was wrong to ask.

EARN TRUST · AIQB Q1 **Q4. "Tell me about a time when you had to communicate a change in direction that you anticipated people would have concerns with. What did you do to understand the concerns and mitigate them?"**

The change in direction was that S/4 would keep the physical book but would no longer be where anyone read "available" — ISM would publish it. Finance heard "you are taking our number away"; the DC heard "a system will tell us what we can pick"; dotcom heard "you will under-promise." I understood the concerns by writing them as rows: what each org needed to read, at what grain, from which document. Finance needed Financial On-Hand to tie to MB5L at month-end — it does, from ISM, at tolerance zero in units, with cost staying S/4-owned. The DC needed the floor to win when it was lower — it does, by the lower-of-two rule. Dotcom needed republish within a minute — the contract is p95 under sixty seconds. The mitigation was the signed table: term, definition, system of record, grain, owner. Three orgs signed it in the FSD. What changed along the way: merchandise reserve got its own visible column because Merchandising's real need was visibility, not promisability.

Number to land: three orgs signed; FOH ties to MB5L at tolerance 0; p95 < 60 s · **Pivot to NACF:** At NACF a definition change — what counts as "received" or "available" — gets a signed table before it gets a system. · **Do NOT say:** the signing orgs' names unless confident (user anchor); the FSD day if unsure — "first week of June."

SAME STORY, THREE ANGLES — DATA · CONFLICT/INFLUENCE/BACKBONE · FAILURE-LEARNING NEW

DATA ANGLE · DIVE DEEP

"One SKU, twelve events, four formulas — and the identity held at 17:00."

Headline metric: Drift = 350 − 350 = 0; Test B = 350

Grain SKU × node × channel × as-of; five feeds, five cadences; the walk shows which events move AFS (commit, STO, floor loss, putaway) and which do not (pick, pack, ship, goods issue). Honest limit: the unit counts are illustrative; the formulas and tolerances are record.

CONFLICT / INFLUENCE ANGLE · HAVE BACKBONE

"The surplus is never promised — 580, not 600."

Headline metric: OOB −20 unpromised; oversell → 0

The room wanted the units; I priced the asymmetry ($30 oversell vs $4.75 undersell, benchmark) and set the phantom probability to one at design time. Leadership accepted visible undersell. Influence: a priced rule in the FSD, not a meeting.

FAILURE / LEARNING ANGLE · ARE RIGHT, A LOT

"My first in-transit rule starved a channel — and the Negative AFS screen could not see it."

Headline metric: P = 0 by default; undersell invisible to a one-sided detector

The master's S4: I first refused to promise against in-transit, then the lane study reversed me (σ ≈ 0.3 day → a ~0.49-day window). Do differently: a watch on unusual undersell beside the Negative AFS report — the detector is blind in that direction by design.

TRAP DRILLS — THE THREE NASTIEST FOLLOW-UPS, WITH NUMERIC ANSWERS NEW

Definitional — is Financial On-Hand before or after reserves? And why isn't Shipped-Not-Billed in it?

FOH includes reserves: Available + Merchandise Reserve + Lost-and-Found = 1,000 + 150 + 0 = 1,150 at the open; a held unit is still on hand. AFS is after reserves and after every committed bucket: 560. SNB is units that have physically left but S/4 still books — they are inside Available until the posting, so FOH carries them already; SNB is shown as its own column so Finance sees what tonight's posting will remove.

canon carries two FOH phrasings — Verify #2

Say: *"FOH is Finance's physical book including reserve; AFS is the promise after reserve and commitments; SNB is the reconciling column between them."*

Definitional — what if S/4 and ISM disagree at T+1?

Test A breaks and it is a Sev until classified. Pull MATDOC lines for that SKU × SLoc with posting time after the watermark — timing, re-cut with MB5B (the +24 row); pull the ISM transition log for a duplicate MBLNR × ZEILE — consumer defect, drop by key and replay (the −10 row); anything left is true drift to Inventory Control with a cause code. Never adjust the book to match the copy.

Say: *"A T+1 difference is a Sev until it has a cause class — timing, duplicate, or true drift — and the book is never edited to make the copy right."*

Definitional — who owns the publish, and what stops two values for one SKU?

ISM's AFS engine is the only publisher, keyed SKU × node × channel × version; the OMS and stores read and never compute; the assortment flag gates dotcom publish. At cutover the rule was published locations per SKU = 1 at every instant — zero on 800 and 900's old publish, then 100% on the new — and Σ AFS was conserved across the switch; rollback triggers (sync < 99% for 4 h, p95 > 2× for 1 h) were armed twice and never fired.

Say: *"One publisher, one key, consumers never compute — and at cutover no SKU was ever published from two places at once."*

20-SECOND VERSION + NACF BRIDGE NEW

20-second version. At Sephora two books each published "available"; I wrote the one signed definition — Basis is the lower of S/4 book and WMS floor, AFS is Basis minus Picking Reserve, Ecom Pick, Ecom Pack and Shipped-Not-Billed — one engine publishes it per SKU, node and channel, consumers never compute, and ISM's copy equals MB52 daily at tolerance zero. Oversell went from recurring to zero through both go-lives; about $2.5M a year, $1M retired support plus roughly $1.5M modeled.

NACF bridge. NACF's equivalent is time-to-available and the identity behind it: receive scan → stow → available, with dock-received units not promisable until stowed (the E1 → E2 window is the dwell side of CAP-D), one published available per ASIN × FC that placement and transfer read, and a zero-tolerance reconciliation between the WMS book and the physical count. I would ask first which system is the book, which is the floor, and what the daily identity between them is.

Safe-say box. "S/4 owns the physical book; ISM publishes the promise; the two are reconciled daily at tolerance zero, never merged." · "Every unit count in my walk is illustrative; the formulas, the tolerance and the result — zero oversell through both go-lives — are the record."

S1-R-infra — Physical truth vs system truth: the infrastructure view of the same program — scan to WMS to S/4 posting, which physical event moves which bucket, what AFS does when a feed or a device is late, and the cutover night from the infrastructure seat NEW

LPs this story serves here: Insist on the Highest Standards (primary — tolerance zero; publish suppressed rather than computed on a stale floor) · Are Right, A Lot (the RF-device script miss, owned) · Bias for Action (stale feed on the cutover wall) · Deliver Results (343 tasks, triggers armed twice, never fired)
Process family: Deliver — physical-to-system latency design, node constraints, cutover runbook and rollback from the infrastructure seat

Honesty label. Record design (FSD / GFiber v8 ISM chapter): WMS on-hand feed ≤ 15 min, late ⇒ publish suppressed for that node, last-good AFS held and flagged; event → republish p95 < 60 s; S/4 movement lines as real-time events; POS lines real-time; S/4 aggregate posting nightly / intraday batch; unmatched > 24 h flag, > 48 h Sev. Record (Apple C2 p24 / resume / SAP Narrative p4): cutover runbook 343 tasks / 48,885 task-minutes ≈ 815 hours; rollback triggers sync < 99% for 4 consecutive hours or order-flow p95 > 2× budget for 1 hour, consumers revert to the prior publish; armed for both go-lives, never fired. Record (Apple G2 p44–45, the S18 RP → JDA WMS program at the same DC family): device test scheduled with engineering days before training; the upgrade script failed on refreshed RF hardware; "hundreds" of devices; 3–4-day estimate; training slipped 1.5 weeks; go-live date held; program landed ~3% under the ~$1.8M budget; mocks 7.0% → 1.8% → 0.3% vs 0.5% tolerance; the what-if rule became the UDC RF-device proof-of-concept before cutover week (Apple G2 p45). Record (Apple D1 p29): 72-hour in-transit freeze; RP read-only 30 days. Illustrative: the 12 EA/hour sell rate, 10 doors × 16 h, 120 UPH, the cutover-night clock times. I owned the latency budgets, the suppression rule, the runbook and the triggers; the WMS, ISM and S/4 engineers built the feeds; DC Ops ran the floor and the devices.

LEADERSHIP PRINCIPLES CARRIED — ONE-LINE PROOF EACH

Insist on the Highest Standards. When the floor feed ages past fifteen minutes the engine holds the last good number and flags the node rather than compute on a lie; drift tolerance is zero.

Are Right, A Lot. I scored a "three-day" device task by effort, not impact; the script failed on new hardware; training slipped a week and a half, the date held, and the what-if rule is permanent.

Bias for Action. On the cutover wall a stale node feed is a re-read, not a re-promise — the rule was decided before anyone was tired.

Deliver Results. 343 tasks, about 815 task-hours, one irreversible instant; rollback armed twice and never fired; zero customer-facing incidents.

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. The availability program lived or died on physical truth reaching system truth in time: an RF scan on the Utah DC floor had to become a WMS status, a WMS status had to reach ISM inside fifteen minutes, an S/4 document had to post as a real-time event, and the published number had to move in under a minute — across hundreds of devices, a dock, a labor plan and a cutover that could not break a peak. **Task.** Own the latency budgets, the rule for what the engine does when physical truth is late, the device and dock dependencies in front of the exit criteria, and the cutover runbook and rollback triggers. **Action.** I wrote the budgets as numbers: floor feed fifteen minutes or publish is suppressed for that node with the last good value held and flagged; event to republish p95 under sixty seconds; store sales real-time, settlement nightly with a twenty-four-hour flag and forty-eight-hour Sev. I mapped every physical event to exactly one bucket transition — receipt, putaway, pick, pack, ship, goods issue, adjustment — so the floor could never move a unit the system did not see. On the earlier WMS migration I had missed a device dependency — a three-day script that failed on new hardware and slipped training a week and a half — so device readiness went into the entrance criteria and an RF-device proof-of-concept ran before cutover week. The cutover was 343 tasks, about 815 hours, inside a transfer freeze, with one irreversible instant — zero-then-publish — and rollback triggers per interface: sync under 99% for four hours or p95 over twice budget for an hour. **Result.** Primary metric: two go-lives, zero customer-facing incidents, rollback armed twice and never fired; a node whose feed went stale under-sold for minutes and never over-sold.

SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

S — Four physical cadences met one published number. A trailer on the dock was in S/4 before it was on the floor; a parcel was off the floor before S/4 knew; a store sale reached S/4 only as a nightly aggregate; a rack leak was on the floor minutes before any document. Hundreds of RF devices, printers and workstations had to run the new code; the dock, labor and cube constraints decided how long the dock-to-stock window was; and a cutover had to move the publish without a SKU ever being promisable from two places.

T — Decide, in numbers, how late physical truth may be before the system stops promising; map every physical event to one bucket; put device and dock readiness in front of the exit criteria; and run the cutover night as a runbook with stop conditions.

Baselines on the table: WMS feed ≤ 15 min · republish p95 < 60 s · aggregate nightly, > 24 h flag, > 48 h Sev · "hundreds" of devices (record) · 343 tasks ≈ 815 h · triggers 99% × 4 h / 2× × 1 h · 72-hour in-transit freeze (S18 lesson) · zero customer-facing incidents allowed.

ACTION — THE PROCESS / OPERATING-MODEL REDESIGN, IN GATED PHASES

- **Discovery (3/29 → 4/30)** — walked the floor with DC Ops: every scan point (receive, putaway, pick, pack, ship, cycle count) and what it posts where; feed cadences and owners; the device fleet and its code level. *Exit bar:* every physical event has one target bucket and one owner; every feed has a cadence and a late rule.

- **Plan & Design (5/15 → 8/14)** — latency budgets written as numbers; the suppression rule (stale floor ⇒ hold last-good, flag the node); the same-cutoff rule for OOB; device readiness and dock readiness as entrance criteria (the G2 what-if rule). *Exit bar:* budgets in the FSD; what-if and fallback per cross-functional dependency.
- **Build & Test (Aug → Jan/Apr)** — the RF-device proof-of-concept before cutover week (record); three mock cutovers against the 0.5% tolerance (7.0% → 1.8% → 0.3%, the S18 arc applied); the replay test; E2E days. *Exit bar:* devices proven, not promised; mock 3 inside tolerance.
- **Cutover & Go-Live (January; May 1 → Jul 1)** — transfer freeze; MCS closing extract · S/4 opening MB52 · ISM seed balanced per bucket (R10); clear negatives; zero-then-publish; OMS re-pointed; triggers armed per interface. *Exit bar:* Σ per bucket equal across the three columns; no SKU published twice; first drift at T+1 = 0.
- **Hypercare / Close (Jun → Jul)** — feed-latency and stale-node flags read on the wall every morning; triggers disarmed only at exit. *Exit bar:* < 2 blockers/day sustained; drift 0 on consecutive days; stale-node flags at zero for a week.

Influence and ownership. Mine: the budgets, the suppression rule, the event-to-bucket map, the entrance criteria, the runbook and the triggers. WMS, ISM and S/4 engineers built the feeds and the jobs; DC Ops owned the floor, the devices and the dock; the war room read the wall.

> **Designed the process; others own it now.** The feed monitors and stale-node flags are Inventory Control's morning read; device readiness is a standing line in every DC runbook. **Trade-off put to leadership:** suppress publish on a stale node (lost sales for minutes) against promising on a floor count that may be wrong; leadership accepted suppression.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY

- **$** — the program protected the S1 headline (about $2.5M a year, split $1M anchor + ≈$1.5M modeled); the S18 migration landed ~3% under ~$1.8M (≈$54K, record / derived). No infrastructure dollar is claimed as savings.
- **#** — 343 tasks; 48,885 task-minutes; one irreversible instant; 2 rollback triggers per interface, armed twice, never fired; "hundreds" of devices proven before training.
- **%** — mocks 7.0% → 1.8% → 0.3% against 0.5% (record, S18); feed completeness design ≥ 99.9%; sync trigger 99%.
- **Time** — floor feed ≤ 15 min; republish p95 < 60 s; dock-to-stock on the worked receipt 100 min (illustrative); training +1.5 weeks with the date held (record).
- **Customer** — zero customer-facing incidents through both go-lives (record).
- **Quality** — a stale node under-sells and never over-sells; every physical event lands in exactly one bucket (record design).

L6 LEVEL EVIDENCE — every line is this story's record or labeled model, no new canon numbers

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	"How late is too late" had no number; I wrote 15 minutes, 60 seconds, 24 hours and 48 hours into the FSD.
Scope & influence	WMS, ISM and S/4 engineering, DC Ops, the device team and Finance — none reported to me.
Execution trade-off + mechanism others own	Suppression over promising on a stale floor; the mechanism others own: the stale-node flag and the feed monitor.
Impact	Two go-lives, zero customer-facing incidents, rollback never fired.
Process improvement	Device readiness moved from a promise to an entrance criterion; the what-if line per dependency is permanent.
Complexity & consensus	Four physical cadences, one published number, a cutover inside a freeze.
Root cause at scale	The device miss root-caused to the risk model (effort vs impact), not to the script.

PROBING — FIVE FACETS, THEN THE HARDEST FOLLOW-UPS NEW

THE FIVE FACETS — WHERE EACH ONE LIVES IN THIS STORY

1 · Data analysis. Grain: event × node × minute. Data: WMS feed as-of timestamps vs ISM receive time (latency distribution per node); ISM publish log (event time → publish time, p95); MATDOC posting time vs watermark; stale-node flags per day. Cuts: by node, by hour, by event type. Revealed: receipts are the slowest physical-to-promisable path (dock-to-stock), goods issues the fastest (document and floor move together); a stale feed hurts only on the floor side — commits keep flowing.

2 · Solution design. Options for a stale floor: (A) keep computing on the last floor count — rejected, promises against a floor that may have lost units; (B) fall back to S/4 book alone — rejected, book is above floor by whatever shipped; (C) hold last-good AFS, flag the node, keep applying commit decrements — chosen: under-sells, never over-sells. Device readiness: promised (rejected after G2) vs proven in an entrance criterion (chosen).

3 · Math & data logic. Exposure = sell rate × stale window; time-to-available = t_putaway − t_101; dock utilization = scheduled unloads ÷ (doors × hours × unload rate); hours = units ÷ UPH × gross-up; trigger exposure = events/hour × (1 − 0.99) × 4 h. Worked below.

4 · Implementation. Phases above; cutover-night runbook (Table 2); hypercare wall. Risks: device code level, feed latency, cutoff mismatch, a 101 across the watermark.

5 · Probe questions. Below: what dotcom sees if the feed dies for an hour; why rollback never fired; why trust the cutover after the device miss; what the dock has to do with AFS.

FACET 5 — PROBE QUESTIONS (CHALLENGE Q&A TUNED TO THIS INTERVIEWER)

Infra — the WMS feed dies for an hour. What does dotcom see?

The last good AFS for that node, flagged stale, still decremented by every commit that arrives — commits come from the OMS, not the WMS. What it cannot see is floor-side change: a putaway (so new stock under-sells until the feed returns) or a loss (so a commit could land on a unit that is gone). The exposure of the second is bounded by sell rate × window — on the worked SKU, 12 an hour × 1 hour = 12 units, and only if a loss happened in that hour. Suppression plus the Negative AFS report at T+1 is the control.

Infra — which physical event moves which bucket?

Receipt 101: S/4 Available up, floor unchanged until putaway — AFS waits. Putaway: floor up — AFS up. Commit: Ecom Pick up — AFS down. STO create: Picking Reserve up — AFS down. Pick/pack confirm: Ecom Pick to Ecom Pack — AFS unchanged. Ship confirm: Ecom Pack to SNB, floor down — AFS unchanged. Goods issue 601/641: S/4 Available down, SNB or Picking Reserve down — AFS unchanged. Cycle count 701/702: Lost/Found — outside AFS; a status change to damaged moves the floor — AFS down instantly.

Infra — what does the dock have to do with AFS?

Time-to-available. A unit is in S/4 at the 101 and promisable only at putaway, so dock-to-stock is a direct AFS lever: on the worked receipt 100 minutes for 240 units. Dock utilization above about 85% makes that window nonlinear, which is why the receive plan and the putaway labor plan are in the latency story, not just the IT story.

Infra — rollback never fired. Were the triggers too loose?

They were priced, not loose: sync under 99% for four consecutive hours — at an illustrative 1,000 events an hour that is more than 10 unapplied events an hour for four hours running — or publish p95 over twice the 60-second budget for a full hour. Both were observable on the wall in real time. They were armed for both go-lives. Not firing means the mocks did their job: 7.0% → 1.8% → 0.3% against 0.5% on the earlier migration, and the same discipline here.

BUSINESS IMPACT MODEL — KPI DRIVERS, BASELINE → AFTER

DRIVER	BASELINE	AFTER	$ / EFFECT	BASIS
Floor-to-system latency rule	none written	≤ 15 min or publish suppressed; last-good held and flagged	oversell exposure bounded by sell rate × window	record design
Publish latency	nightly / batch dialects	event → republish p95 < 60 s	stale-promise leak priced by the 60-s budget	record design
Device readiness	promised (3–4-day estimate)	proven in entrance criteria; PoC before cutover week	training +1.5 weeks once; never since	record
Cutover	—	343 tasks ≈ 815 h; triggers armed twice, never fired	≈$90K runbook effort at an illustrative $110/h	record / illustrative
Mock reconciliation	7.0%	0.3% vs 0.5% tolerance	—	record (S18)

THE MATH — FOUR-TABLE DERIVATION (FORMULA → SUBSTITUTION → RESULT) NEW

Table 1 — before: the latency problem in numbers (physical event → system truth).

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Time-to-available on the worked receipt	`t_putaway − t_101 = 09:10 − 07:30`	illustrative (S1-R E1 → E2)	100 min; 240 EA unpromisable
2	Floor-feed stale exposure	`sell rate × window = 12 EA/h × 15 min = 12 × 0.25`	illustrative sell rate; record 15-min budget	3 EA
3	Same exposure with a 1-hour outage and no suppression	`12 × 1.0`	illustrative	12 EA; $ = 12 × $30 = $360
4	Store-sale phantom window (no SNB)	`W = t_post − t_sale; a 10:02 sale posts at the nightly aggregate ≈ 14 h; exposure ≈ sales rate × W`	record formula (S3); illustrative hours	≈14 h per sale line
5	Dock utilization (receive plan)	`scheduled ÷ (doors × hours × rate) = 130 ÷ (10 × 16 × 1.0)`	booklet CAP-D formula; illustrative inputs	81.3% (< ~85%)
6	Putaway labor on the receipt	`hours = units ÷ UPH × gross-up = 240 ÷ 120 × 1.2`	booklet CAP-B formula; illustrative inputs	2.4 h
7	Device dependency as planned	`effort estimate 3–4 days; what-if plans = 0`	record (Apple G2)	single-point dependency

Table 2 — the fix: latency rules, device gate and the cutover-night runbook, values substituted.

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Suppression rule	`if (now − WMS as-of) > 15 min: publish(node) = last-good, flag = stale; commits still apply`	record design	oversell exposure from floor loss → 0 beyond 15 min
2	Republish budget	`p95(t_publish − t_event) < 60 s`	record design	60 s
3	Settlement budget	`unmatched age > 24 h → flag; > 48 h → Sev`	record design	24 h / 48 h
4	Device gate	`devices proven ÷ devices in fleet = 100% before training entrance; PoC before cutover week`	record (Apple G2 p45)	proven, not promised
5	Runbook size	`48,885 task-min ÷ 60 = 814.75 h over 343 tasks → 142.5 min per task average`	record / derived	≈815 h; ≈2.4 h/task
6	Cutover night (illustrative clock, record steps)	`T−72 h freeze in-transit → 20:00 MCS closing extract by bucket → 21:30 S/4 opening MB52 → 22:00 ISM seed (opening buckets incl. unposted ship confirms = opening SNB) → 22:30 R10: Σ per bucket across three columns = 0 difference → 23:00 clear negatives → 23:30 transfer AFS to Uncommitted → 00:30 scan-and-reserve → 01:00 zero-then-publish (irreversible) → 01:05 OMS reads ISM → triggers armed → 06:00 first drift`	record steps (Apple D1 freeze; GFiber R10; Packet 3 U5); clock illustrative	one irreversible instant
7	Trigger 1 exposure	`events/h × (1 − 0.99) × 4 h = 1,000 × 0.01 × 4`	record trigger; illustrative volume	> 40 unapplied events over 4 h fires rollback
8	Trigger 2	`p95 > 2 × 60 s = 120 s for 60 consecutive min`	record trigger	120 s × 1 h
9	Double-publish invariant	`published locations per SKU at any instant = 1; Σ AFS(before) = Σ AFS(after)`	record (R10; U5)	0 double publish

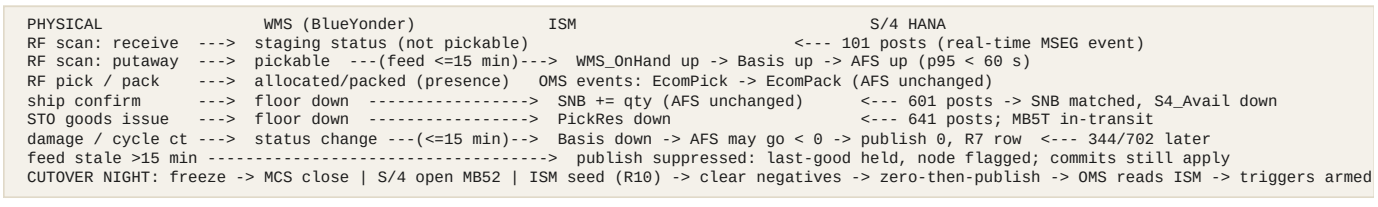
Table 3 — after: the metric, before → after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	Stale-floor behavior	compute on last count	hold last-good, flag node, commits still apply	record design
2	Device readiness	promised; script failed; training +1.5 weeks	entrance criterion; PoC before cutover week; no repeat	record
3	Mock reconciliation	7.0%	1.8% → 0.3% (vs 0.5%)	record (S18)
4	Rollback	—	armed twice, never fired	record
5	Customer-facing incidents	recurring oversell	0 through both go-lives	record
6	Go-live date	at risk from the device slip	held	record

Table 4 — the dollars, every multiplication shown.

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Runbook effort	`814.75 h × $110 = $89,622.50`	record hours; illustrative rate	≈$90K
2	S18 budget outcome	`$1,800,000 × 3% = $54,000 under`	record ~3%; derived $	≈$54K
3	Stale-hour exposure without suppression	`12 EA × $30 = $360 per SKU-hour of floor loss`	illustrative	$360
4	S18 pipeline behind the date	`$20M ÷ 8 months = $2.5M/month average`	record $20M; derived division — "not banked"	≈$2.5M/month (the S18 noun)
5	Protected headline (S1)	`$1,000,000 anchor + $1,530,000 modeled`	user anchor + modeled	≈$2.5M/yr (the S1 noun)

SOLUTION FLOW — THE SYSTEM & DATA MAP (LATENCY VIEW)



SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Sell rate on the SKU (EA/h) × 15-min window	4 → 1 EA	12 → 3 EA	40 → 10 EA

INPUT	LOW	BASE	HIGH
Feed budget	5 min → 1 EA at base rate	15 min → 3 EA	60 min → 12 EA
Dock utilization (130 trailers)	12 doors → 67.7%	10 doors → 81.3%	8 doors → 101.6% (dwell nonlinear)
Putaway UPH on 240 EA (×1.2)	200 → 1.44 h	120 → 2.4 h	60 → 4.8 h
Trigger 1 at 99% × 4 h	500 ev/h → 20	1,000 → 40	5,000 → 200 unapplied events

SPOKEN STATEMENT

"From the infrastructure seat the program was four physical cadences meeting one published number. I wrote the budgets as numbers: the floor feed inside fifteen minutes or the engine holds the last good value and flags the node — it will under-sell for minutes and never over-sell; republish within sixty seconds; store settlement nightly with a day's flag and two days to a Sev. Every physical event maps to one bucket — a receipt waits for putaway, a commit moves the number in a minute, a pick, pack or ship moves units between buckets and leaves it alone, a loss on the floor moves it instantly. I had missed a device dependency on the earlier migration — a three-day script that failed on new hardware and slipped training a week and a half, date held — so device readiness became an entrance criterion and a proof-of-concept ran before cutover week. The cutover was 343 tasks, about 815 hours, one irreversible instant, rollback triggers armed twice and never fired, zero customer-facing incidents."

IF CHALLENGED

"The device story is a different program — why is it here?"

Because it is the same DC family and the lesson is the record of how the ISM cutover was planned: Apple G2 says the what-if rule became the UDC RF-device proof-of-concept before cutover week. I keep the programs separate — the miss is S18's, the proof-of-concept is UDC's — and I say so.

"Your clock times for cutover night are invented."

The clock is illustrative; the steps and their order are record — freeze, three-column balance, clear negatives, transfer to Uncommitted, scan-and-reserve, zero-then-publish, triggers armed, first drift at T+1. I would not quote the hours.

TWIST MAP — LIKELY LP QUESTIONS FROM THIS INTERVIEWER → THE ANGLE TO TAKE

LIKELY QUESTION (THIS INTERVIEWER)	LP	ANGLE — WHAT TO EMPHASIZE	SKIP / NEVER SAY
What happens when the floor and the system disagree?	Insist on the Highest Standards	Suppress, hold last-good, flag; tolerance zero at T+1.	Never "we trusted the WMS."
Tell me about a go-live that went wrong.	Are Right, A Lot	Device script: effort vs impact; +1.5 weeks; date held; permanent rule.	Never "100 of 400"; never a cost multiplier.
Tell me about acting fast with incomplete information.	Bias for Action	Stale node on the wall: re-read, not re-promise — rule pre-decided.	Do not dramatize; it was a rule firing.
How do you run a cutover?	Deliver Results	343 tasks, one irreversible instant, priced triggers, never fired.	Do not quote the clock times as record.

BEHAVIORAL PACK — FOUR SEPARATE 90-SECOND STAR ANSWERS, EACH FROM A DIFFERENT MOMENT OF THE STORY

NEW

INSIST ON THE HIGHEST STANDARDS · AIQB Q1 **Q1. "Describe a time when you refused to compromise your standards around quality/customer service, etc. Who was your customer? What was the result?"**

The compromise on offer was to keep publishing availability from the last warehouse count when the floor feed went stale. The customer was the dotcom shopper who gets a "yes" we then cannot ship. I wrote the rule the other way: if the floor feed is older than fifteen minutes, the engine holds the last good number for that node, flags it, and keeps applying order commits — so it can under-sell for a few minutes and can never over-sell against a floor that may have lost units. The cost is lost sales during a stale window — on a SKU selling twelve an hour that is three units per fifteen minutes — and more flags for Inventory Control to read. The same posture is the drift tolerance: zero, a difference between books is a Sev. The result is two go-lives with zero customer-facing incidents and a stale node that under-sold for minutes and never over-sold.

Number to land: 15 min; 3 EA per stale window at 12/h (illustrative); drift 0 · **Pivot to NACF:** I would hold the same line between receive scan and available: not stowed, not promisable, whatever the dashboard says. · **Do NOT say:** that suppression was free; that the sell rate is program data.

ARE RIGHT, A LOT · AIQB Q4 **Q2. "We don't always make the right decision all the time. Tell me about a time when you made a bad decision. What was the impact of that decision? What did you learn?"**

On the WMS migration I planned a device conversion as a three-to-four-day task, low effort, low attention — and that was the bad decision. Training sign-off was a go-live exit criterion gated on hundreds of RF guns, printers and workstations converting. I scheduled a device test with engineering days before training just to be sure, and the upgrade script failed: the fleet had new RF devices that did not run the old code. With key people on night shifts, manual conversion could not finish in time. I told leadership straight: training slips a week and a half, my miss. The root cause was not the script; it was my risk model — I had scored the task by effort, not impact, so a single-point dependency in front of an exit criterion had no what-if plan. The rule changed permanently: every cross-functional dependency gets a what-if and a fallback regardless of effort, and device readiness moved into training's entrance criteria — proven devices, not promised ones. The go-live date held, the program landed about three percent under its $1.8M budget, and on the next DC an RF-device proof-of-concept ran before cutover week.

Number to land: +1.5 weeks; date held; ~3% under ~$1.8M · **Pivot to NACF:** FC launches are chains of small dependencies — a label printer, a firmware level, a dock appointment system — impact decides attention, never effort. · **Do NOT say:** "100 of 400" (sources disagree — say "hundreds"); any "x cheaper" multiplier.

BIAS FOR ACTION · AIQB Q4 **Q3. "Tell me about a time when you had to gather information and respond immediately to a situation. What was the outcome?"**

On the cutover wall a node's floor feed aged past fifteen minutes while orders were flowing. The information I needed was already on the wall because we had designed it to be there: the feed as-of time per node, the stale flag, the publish log, and the drift and OOB rows from the last cut. The engine had already done the first thing — suppressed publish for that node and held the last good number. The decision in the room was whether to re-promise from the S/4 book instead; I said no, because the book is above the floor by whatever shipped since the last snapshot, and promising from it is exactly the oversell we had designed out. We re-read when the feed returned, let the Negative AFS report tell us if any commit had landed on lost stock — none had — and logged the event. The outcome was a few minutes of under-sell on one node, no customer impact, and a runbook line for next time. The speed came from deciding the rule when nobody was tired.

Number to land: 15-min budget; exposure ≤ sell rate × window; 0 negatives at T+1 · **Pivot to NACF:** On a launch night the decision rules — what we hold, what we never promise — should be written before the first trailer. · **Do NOT say:** that I overrode the engine; that the episode's duration is record.

DELIVER RESULTS · AIQB Q2 **Q4. "Tell me about a time when you had significant, unanticipated obstacles to overcome in achieving a key goal. Were you eventually successful?"**

The goal was two go-lives — January planning, May execution — with zero customer-facing incidents and a peak freeze behind them; the obstacle was that cutover moved a live publish across about a dozen impacted systems with one instant that could not be undone. I ran it as a 343-task runbook, about 815 task-hours, each task with an owner, a duration and a predecessor, inside a transfer freeze so in-transit converted as its own batch. The night had a three-column balance — MCS closing, S/4 opening, ISM seed — that had to agree per bucket before anything moved; then clear negatives, transfer to Uncommitted, scan-and-reserve, and zero-then-publish so no SKU was ever promisable from two places. Rollback triggers were numbers per interface — sync under 99% for four consecutive hours or publish p95 over twice budget for an hour — armed for both go-lives. They never fired. First drift at T+1 was zero; hypercare exited at under two blockers a day in week three of four. Successful, and the obstacle I would plan harder for next time is the device fleet.

Number to land: 343 / ≈815 h / armed twice, never fired / drift 0 at T+1 · **Pivot to NACF:** An FC go-live runbook with one named irreversible step and priced stop conditions is the same artifact. · **Do NOT say:** cutover clock times as record; "the 836-row log" as ISM's — it is the umbrella program's.

SAME STORY, THREE ANGLES — DATA · CONFLICT/INFLUENCE/BACKBONE · FAILURE-LEARNING NEW

DATA ANGLE · DIVE DEEP

"Latency per node, per hour, per event type — receipts slowest, goods issues fastest."

Headline metric: ≤ 15 min feed; p95 < 60 s; 100-min dock-to-stock (illustrative)

Event × node × minute grain; the stale-node flag and publish log were the data. Revealed: a stale feed only hurts on the floor side; commits keep flowing. Honest limit: the sell rate and clock are illustrative.

CONFLICT / INFLUENCE ANGLE · HAVE BACKBONE

"Suppress publish on a stale node — and accept the lost sales."

Headline metric: under-sell ≤ 3 EA per window vs oversell $30/unit

Dotcom wanted the number to keep moving; I priced the asymmetry and leadership accepted suppression. Influence: a rule in the FSD and a flag on the wall, not a debate at 01:00.

FAILURE / LEARNING ANGLE · ARE RIGHT, A LOT

"Effort is not impact."

Headline metric: training +1.5 weeks; date held

The device script (Apple G2) — scored by effort, no what-if. Do differently: what-if per dependency from kickoff; device compatibility verified at refresh. Proof it stuck: UDC RF-device PoC before cutover week; mocks 7.0 → 1.8 → 0.3%.

TRAP DRILLS — THE THREE NASTIEST FOLLOW-UPS, WITH NUMERIC ANSWERS NEW

Definitional — the ship confirm and the 601 are minutes apart. In that gap, where is the unit?

Off the floor, inside S/4 Available, and inside SNB at the DC node. Floor 1,090 + SNB 130 = 1,220 = the netted floor; S/4 still 1,240; Basis 1,220; buckets 550; AFS 670 — unchanged. When the 601 posts, S/4 goes to 1,150 and SNB to 40 together; AFS still 670. The unit is never in two promisable places and never in none.

Say: *"Between ship confirm and goods issue the unit sits in Shipped-Not-Billed — subtracted from the promise, still in Finance's book — and AFS does not move."*

Why "hundreds" of devices — don't you know the number?

Two of my own source documents disagree — one says about 400, the other about 100 — so I say "hundreds" and the estimate was three to four days. The count is not the lesson; the missing what-if plan is.

Say: *"Hundreds of devices, a three-to-four-day estimate, and no what-if plan — that last part was mine."*

Your triggers never fired — how would you know they worked?

They were exercised in the mocks against the 0.5% tolerance and the sync and p95 series were on the wall in real time; a trigger that is observable and armed is a working control even when it does not fire. If I had seen sync under 99% for four hours, consumers would have reverted to the prior publish — a tested path, not a hope.

Say: *"Armed and observable for both go-lives; never needed — that is the outcome I wanted, and the mocks are why."*

20-SECOND VERSION + NACF BRIDGE NEW

20-second version. From the infrastructure seat the availability program was four physical cadences meeting one published number: floor feed inside fifteen minutes or publish is held and flagged, republish under a minute, every physical event mapped to one bucket, devices proven before training after I missed one, and a 343-task cutover with one irreversible instant and rollback triggers armed twice and never fired — zero customer-facing incidents.

NACF bridge. NACF's version is dock-to-stock as the promisability lever — dock doors × hours × unload rate capped near 85% utilization, putaway hours = units ÷ UPH — and a rule that received-not-stowed is never promisable; plus a launch runbook with device readiness in the entrance criteria and stop conditions written as numbers before the first trailer.

Safe-say box. "A stale floor feed means we hold the last good number and flag the node — we under-sell for minutes and never over-sell." · "The device miss is the earlier WMS migration's; the proof-of-concept before cutover week is how it changed the next one — hundreds of devices, a week and a half on training, the date held."

S3-R — End to end: Shipped-Not-Billed and the one Financial On-Hand — a store sale that S/4 will not see until tonight, de-aggregated back to the line and the cent, so Finance and Supply read one on-hand and the failure direction is always safe NEW

LPs this story serves here: Earn Trust (primary — Finance keeps its own reconciliation → Finance owns the aging report) · Customer Obsession (decrement at the sale, not at the posting) · Ownership (the product is Finance running it without me) · Dive Deep (match rate and aging as the metric that proves it)
Process family: Deliver / Finance — sold-not-posted control: SNB ledger, aggregate-posting match, COGS de-aggregation, one Financial On-Hand, month-end tie

Honesty label. Record mechanism (GFiber v8 S3, Amazon master S3): SNB incremented by POS flash sale, OMS omni pending sale and WMS ship confirm; decremented only on a matched S/4 aggregate posting; match Aggregation# ↔ store header ↔ lines; park unless Σq_i = q_P; duplicate header dropped by key; > 24 h flag, > 48 h Sev; COGS_i = C_P × q_i ÷ q_P with the last line absorbing rounding so Σ ties to the cent; SNB ≥ 0 always; FOH published by ISM, ties to MB5L at month-end; cost (WAC) stays S/4-owned, ISM visibility-only. User anchor (GFiber v8 only): "Financial On-Hand truths 2 → 1"; FSD signed 6/5/25; match rate ≥ 99.5% within 24 h is a *design target*, not a result; days-to-tie at month-end not in record. Modeled: ≈$0.6M/yr (cancels $450,000 + reconciliation $141,440 + close $48,960 = $640,400) — "the record has no drift dollar"; the cancel-rate row borrows S1's 0.10% benchmark, so if the room wants one number keep S1's and drop it. Illustrative: the store 0101 walk (60 → 48), the 12-unit / $300.00 posting, the $100.00 three-line split. I owned the identity, the match rule, the aging tolerance and the audit path; the ISM and S/4 teams built the feeds, the matcher and the screens; Inventory Accounting runs the aging report.

LEADERSHIP PRINCIPLES CARRIED — ONE-LINE PROOF EACH

Earn Trust. Finance said "we keep our own reconciliation"; I showed them the same on-hand arriving from the physical side first and gave them the aging report to own — two truths became one.

Customer Obsession. A store sale S/4 would not see until the nightly aggregate was a unit dotcom could still promise; I moved the decrement to the moment of sale.

Ownership. The exit bar was not "the matcher works" but "Inventory Accounting runs the unmatched-aging report without me."

Dive Deep. The failure direction is provable: an unmatched posting keeps SNB high, so AFS_reported ≤ AFS_true in every unmatched state — under-sell, never over-sell.

60-SECOND BUSINESS-FIRST PITCH — STOP ON THE PRIMARY METRIC

Situation. S/4 posts store sales the way an ERP should — as a nightly aggregate, one quantity and one cost per aggregation number — while the stores sell all day. In that window a sold unit was still "available" to dotcom, and Finance and Operations each kept their own on-hand because neither trusted the other's timing. **Task.** Make the sold-but-not-posted units visible to the promise in real time, tie the aggregate posting back to the lines and the cents, and publish one Financial On-Hand both orgs read. **Action.** I defined Shipped-Not-Billed as its own bucket: incremented at the store ring, the omni pending sale and the DC ship confirm, keyed so a replay is a no-op; decremented only when the S/4 aggregate posting matches — aggregation number to store header to lines, parked unless the line quantities sum to the posting quantity. The matcher pro-rates the posting's cost back to each line — cost times line quantity over posting quantity, last line absorbs the rounding, so a $300.00 posting over twelve units in five lines is 75, 50, 100, 25, 50 to the cent. Financial On-Hand is published once from ISM — Available plus merchandise reserve plus lost-and-found, with SNB as its own column — and ties to MB5L at month-end; cost stays S/4-owned. The failure direction is safe by construction: a posting that will not match keeps SNB high and under-states the promise. **Result.** Primary metric: Financial On-Hand truths two to one, Finance reading the number from ISM and owning the aging report; over-sell from unmatched postings zero by design; about $0.6M a year modeled, record carries no drift dollar.

SITUATION / TASK — THE BUSINESS PROCESS PROBLEM, SCOPE, SCALE, BASELINES

S — A store rings three units at 10:02; S/4 learns of it at the nightly aggregate. Until then the book says the units are on hand, dotcom's old computation said they were promisable, and Finance's bridge spreadsheet tried to reconcile what Operations counted against what S/4 posted. Two Financial On-Hand truths, a phantom-availability window the length of a business day, and a month-end that closed on a bridge.

T — Give sold-not-posted units a bucket that moves at the sale and clears only on a matched posting; de-aggregate the posting to the line and the cent for the data platform; publish one FOH that Finance accepts and that ties to MB5L; hand the exception report to Finance to own.

Baselines on the table: FOH truths 2 · phantom window W = t_post − t_sale ≈ a business day (illustrative) · Finance bridge ≈ 32 h/week (benchmark) · month-end tie-out +1 day (benchmark) · match-rate design target ≥ 99.5% within 24 h · SNB ≥ 0 always.

ACTION — THE PROCESS / OPERATING-MODEL REDESIGN, IN GATED PHASES

- **Discovery** — where every sold-not-posted unit comes from (POS flash sale, omni pending sale, ship confirm) and how S/4 posts it (aggregate, Aggregation#, qty, COGS); who reads on-hand and from where. *Exit bar:* every sold-not-posted path named; Finance's bridge spreadsheet documented as the baseline.

- **Plan & Design (inside the ISM FSD chain; signed 6/5/25 — user anchor)** — SNB ledger definition; match rule; de-aggregation rule; failure-direction statement; FOH equation; control owner per bucket. *Exit bar:* Finance accepts one FOH; WAC confirmed S/4-owned, ISM visibility-only.
- **Build & Test** — matrix families: SNB increment/decrement/reversal, de-aggregation, duplicate header, late line; Inventory Accounting's UAT entrance "every SIT case ties to the cent." *Exit bar:* Σ per-line COGS = posting COGS to the cent in every cell; SNB never negative in any cell.
- **Cutover & Go-Live (January; May)** — opening SNB seeded from unposted ship confirms (R10); publish FOH and SNB back to S/4 visibility. *Exit bar:* aging report empty or explained at T+1; no negative SNB; Finance reads FOH from ISM.
- **Hypercare / Close** — aging report read daily; month-end FOH tie to MB5L through the ISM number, not a bridge. *Exit bar:* aging report owned by Finance without me; month-end ties (mechanism canonical; days-to-tie not in record).

Influence and ownership. Mine: the identity, the match rule, the aging tolerance, the audit path, the UAT entrance bar. ISM and S/4 teams built the feeds, matcher and screens; Inventory Accounting owns the aging report and the settlement side; Store Systems own the POS feed.

> **Designed the process; others own it now.** Inventory Accounting runs unmatched-aging; the EDW carries per-line COGS with Aggregation# lineage. **Trade-off put to leadership:** a posting that will not match under-states the promise until it is fixed (visible over-ordering by Replenishment) — accepted, because the other direction is a customer defect.

RESULTS — $ · # · % · TIME · CUSTOMER · QUALITY

- **$** — ≈$0.6M/yr modeled: cancels 30,000,000 × 0.10% × $15 = $450,000; reconciliation 32 h/wk × 52 × $85 = $141,440; close 6 people × 8 h × $85 × 12 = $48,960; total $640,400. The record has no drift dollar.
- **#** — FOH truths 2 → 1 (user anchor); 3 increment sources, 1 decrement path; 5 lines / 12 units / $300.00 tie to the cent (illustrative).
- **%** — match-rate design target ≥ 99.5% within 24 h; over-sell from unmatched postings 0% by construction.
- **Time** — AFS decrement moved from t_post (nightly) to t_sale (real-time); > 24 h flag, > 48 h Sev; month-end tie through ISM.
- **Customer** — the phantom-availability window closed: a unit sold at 10:02 is off the promise at 10:02.
- **Quality** — SNB ≥ 0 always; Σ per-line COGS = C_P to the cent; FOH_ISM − FOH_S4 = 0 at month-end per SKU × node (design).

L6 LEVEL EVIDENCE — every line is this story's record or labeled model, no new canon numbers

L6 DIMENSION	EVIDENCE IN THIS STORY
Ambiguity	Two orgs, two on-hands, no shared definition of "sold"; I defined the bucket and the match.
Scope & influence	Finance, Inventory Accounting, Store Systems, Digital and the ISM/S/4 teams; the FSD signature was the lever.
Execution trade-off + mechanism others own	Under-state on unmatched rather than guess; the mechanism others own: the aging report.
Impact	FOH truths 2 → 1; Finance closes on the ISM number; phantom window closed.
Process improvement	Bridge spreadsheet retired by giving its owner the real report.
Complexity & consensus	Three increment sources, one decrement path, cents that must tie, two orgs that had to agree.
Root cause at scale	Drift surfaces on the aging report, never inside the FOH number — late line, header-as-lines, return without original.

PROBING — FIVE FACETS, THEN THE HARDEST FOLLOW-UPS NEW

THE FIVE FACETS — WHERE EACH ONE LIVES IN THIS STORY

1 · Data analysis. Grain: SKU × node × sales-event line, rolled to Aggregation# for the match; FOH at SKU × node × day. Data: Aptos RetailSaleTransactions (Txn × line × store), OMS Tlog / ShipmentConfirmation lines, S/4 aggregate posting (Aggregation#, qty, COGS; stock side 601 in MATDOC, FI side ACDOCA). Cuts: unmatched by age, by cause (late line, header-as-lines, return without original), by store. Revealed: the phantom window was cadence, not error — so the fix is where the decrement happens, not a faster batch.

2 · Solution design. Options: (A) faster S/4 posting — rejected, still a window and S/4 posts aggregates by design; (B) let dotcom read POS directly — rejected, a second computation; (C) an SNB bucket inside the one AFS identity, cleared only on a matched posting, with FOH published once — chosen. Inside C: partial-match vs park-unless-equal — park chosen (failure direction safe).

3 · Math & data logic. SNB(t) = Σ sold-or-shipped-not-posted − Σ matched; COGS_i = C_P × q_i ÷ q_P, last line absorbs rounding; W = t_post − t_sale; AFS_reported ≤ AFS_true for all unmatched states; match_rate = matched ÷ received per day. Worked below on store 0101.

4 · Implementation. Phases above; opening SNB seeded at cutover from unposted ship confirms; Finance UAT entrance "ties to the cent"; aging report handoff. Risks: late POS lines, duplicate headers, returns without originals.

5 · Probe questions. Below: why isn't SNB in on-hand; what if S/4 posts 12 and ISM saw 10 lines; why quantity pro-rata; who owns cost; and the trap drills.

FACET 5 — PROBE QUESTIONS (CHALLENGE Q&A TUNED TO THIS INTERVIEWER)

Infra — why isn't Shipped-Not-Billed in on-hand?

It is in Finance's book and out of the promise. Until the posting, S/4 Available still contains the sold unit — so Financial On-Hand, which starts from Available, already carries it; SNB is shown as its own column so Finance sees what tonight's posting will remove. AFS subtracts SNB because the unit is gone from the promise the moment it is sold. After the match, Available falls by the matched quantity and SNB falls by the same — the two books agree and SNB is back to zero.

> **Infra — S/4 posts 12 units; ISM has seen only 10 lines. What happens?**
> Park the posting — the rule is match only when Σq_i = q_P. SNB stays at 10, S/4 Available has already dropped by 12, so AFS is under-stated by 10 until the two lines land; the unmatched-aging report shows it at 24 hours and makes it a Sev at 48. I never partial-match to force a zero, because the two missing lines may be a different store's or a return without its original.

> **Infra — why pro-rate cost by quantity? Lines can carry different prices.**
> Price, yes; cost, no — the posting is per SKU, so every unit in it carries the same cost and quantity pro-rata is exact cost pro-rata. Price lives on the sales line, not the posting. The only arithmetic risk is rounding, and the last line absorbs it so the sum ties to the cent.

> **Infra — who owns cost?**
> S/4. Moving-average cost (MBEW-VERPR) is S/4's; ISM displays it and multiplies it for the MB5L tie. A value difference at month-end is a WAC question for Finance, never an ISM edit. Tolerance on units is zero.

BUSINESS IMPACT MODEL — KPI DRIVERS, BASELINE → AFTER

DRIVER	BASELINE	AFTER	$ / YR	BASIS
Financial On-Hand truths	2 (Finance vs Ops)	1 shared FOH	—	user anchor (FSD 6/5/25)
AFS decrement point	at S/4 posting	at sale / ship	—	record mechanism
Phantom-availability cancels	~0.10 pt of orders	~0	≈$0.45M	modeled (30M × 0.10% × $15)
Finance reconciliation hours	32 h/wk bridge	~0	≈$0.14M	modeled (32 × 52 × $85)
Month-end close, inventory tie-out	+1 day bridge	ties to MB5L	≈$0.05M	modeled (6 × 8 h × $85 × 12)
Total modeled	—	—	≈$0.6M/yr	record has no drift dollar

THE CALCULATION WALK — ONE STORE, ONE SKU, ONE DAY, ONE POSTING NEW

Setup (illustrative). SKU 12345 at store 0101 (no WMS floor, so Basis = S4_Avail; no MerchRes at a store). Opening after the 06:00 snapshot: S/4 Available 60; SNB 0; AFS 60. Five sales during the day — 3, 2, 4, 1, 2 units — one nightly aggregate posting P with q_P = 12 and C_P = $300.00.

Table W1 — the day at the store: ring → SNB → AFS, then the posting.

#	TIME · EVENT	S4_AVAIL	BASIS	SNB	AFS = BASIS − SNB	NOTE
F0	06:00 snapshot	60	60	0	60 − 0 = 60	store: Basis = S4_Avail
F1	10:02 POS rings 3 EA (Kafka line, key POS × Txn × line)	60	60	0 + 3 = 3	60 − 3 = 57	republished < 60 s
F2	10:04 the same transaction replays	60	60	3 (no-op by key)	57	R9: duplicate dropped
F3	11:15 · 13:40 · 15:05 · 16:30 rings of 2, 4, 1, 2	60	60	3 + 2 + 4 + 1 + 2 = 12	60 − 12 = 48	physical on shelf = 48
F4	17:00 watermark (pre-posting): Finance reads FOH columns	60	60	12	48	Avail 60 · SNB 12 · physical 48 — one row, both orgs
F5	23:30 S/4 aggregate posting P: Aggregation# A-1, q_P = 12, C_P = $300.00	60 − 12 = 48	48	12 − 12 = 0 (Σq_i = 12 = q_P → match)	48 − 0 = 48	AFS unchanged across the posting
F6	T+1 06:00 drift	48 = MB52 48	48	0	48	Drift = 48 − 48 = 0

Table W2 — de-aggregation to the cent (COGS_i = C_P × q_i ÷ q_P; last line absorbs rounding).

LINE	Q_I	FORMULA → SUBSTITUTION	COGS_I	RUNNING Σ
1	3	300.00 × 3 ÷ 12	75.00	75.00
2	2	300.00 × 2 ÷ 12	50.00	125.00
3	4	300.00 × 4 ÷ 12	100.00	225.00
4	1	300.00 × 1 ÷ 12	25.00	250.00
5 (last)	2	300.00 − 250.00 (absorbs rounding; nominal 50.00)	50.00	300.00 ✓ = C_P
Rounding case A	1, 1, 1	C_P = $100.00: 100 ÷ 3 = 33.333… → 33.33, 33.33, last = 100.00 − 66.66	33.33 · 33.33 · 33.34	100.00 ✓
Rounding case B	3, 2, 2	C_P = $100.00: 100 × 3 ÷ 7 = 42.857 → 42.86; 100 × 2 ÷ 7 = 28.571 → 28.57; last = 100.00 − 71.43	42.86 · 28.57 · 28.57	100.00 ✓

Table W3 — what goes wrong, one variant per control.

CONTROL	VARIANT	FORMULA → SUBSTITUTION	CAUGHT BY · OWNER	EFFECT
R8 park rule	One 2-unit line has no header key; 10 of 12 match	`matched 10 → SNB 12 − 10 = 2; S4_Avail 48; AFS = 48 − 2 = 46 vs truth 48`	unmatched-aging > 24 h · Inventory Accounting	−2 (under-sell, safe)
R8 park rule (whole posting)	ISM has seen 10 lines; S/4 posts 12	`Σq_i 10 ≠ q_P 12 → park; SNB 10; S4_Avail 48; AFS = 48 − 10 = 38 vs truth 48`	aging; Sev at 48 h	−10 (safe)
R9 duplicate header	Posting A-1 arrives twice	`match set already closed → rejected; SNB stays 0; a decrement beyond matched quantity is rejected → SNB never −12`	R9 · ISM app team	0
SNB ≥ 0	A return line lacks its original	`decrement attempt on SNB 0 → rejected and alerted; line to aging report`	R3/R8	0
Cents tie	Naive rounding on case A	`33.33 × 3 = 99.99 ≠ 100.00 → UAT cell fails; last-line rule → 100.00`	Inventory Accounting UAT entrance	0.01 caught
R11 month-end	Units tie, value does not	`Σ FOH_qty × MBEW-VERPR ≠ MB5L value with qty equal → WAC question (S/4 owns), not an ISM edit`	R11 · Finance	units 0; value → Finance

THE MATH — FOUR-TABLE DERIVATION (DOLLARS; FORMULA → SUBSTITUTION → RESULT) NEW

Table 1 — before.

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	FOH truths	`Finance book + Ops count`	user anchor	2
2	Phantom window	`W = t_post − t_sale = 23:30 − 10:02`	illustrative	≈13.5 h
3	Cancels from phantom availability	`30,000,000 × 0.10% × $15`	modeled; CS $5–8 + refund + margin share	$450,000
4	Finance bridge labor	`32 h/wk × 52 × $85`	benchmark	$141,440
5	Month-end tie-out	`6 × 8 h × $85 × 12`	benchmark	$48,960

Table 2 — the fix.

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Decrement point	`AFS(10:02) = 60 − 3 = 57 (was 60 until 23:30)`	record mechanism	W → 0
2	Match	`Σq_i = 3 + 2 + 4 + 1 + 2 = 12 = q_P → SNB 12 → 0; S4_Avail 60 → 48`	record rule	48 = 48
3	De-aggregation	`75 + 50 + 100 + 25 + 50 = 300.00 = C_P`	record rule	ties to the cent
4	Match-rate KPI	`matched ÷ received per day ≥ 99.5% within 24 h; at 2,000 postings/day → ≤ 10 unmatched`	design target	≤ 10/day
5	Failure direction	`unmatched ⇒ SNB high ⇒ AFS_reported ≤ AFS_true`	record statement	under-sell only

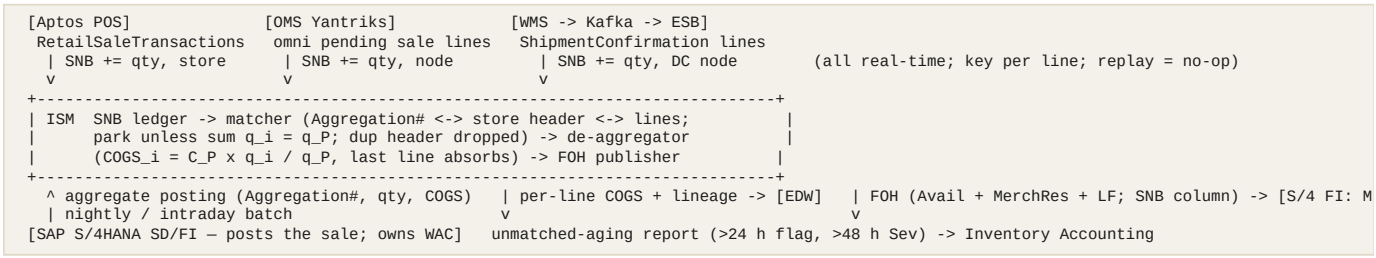
Table 3 — after.

STEP	METRIC	BEFORE	AFTER	BASIS
1	FOH truths	2	1	user anchor
2	Decrement point	t_post	t_sale	record
3	Over-sell from unmatched postings	possible	0 by construction	record design
4	Finance bridge	32 h/wk	~0; aging report owned by Finance	benchmark / record
5	Month-end	bridge, +1 day	ties to MB5L through ISM	record mechanism; days not in record

Table 4 — the dollars.

STEP	QUANTITY	FORMULA → SUBSTITUTION	BASIS	VALUE
1	Cancels avoided	`30,000,000 × 0.001 × $15 = $450,000`	modeled	$0.45M
2	Reconciliation labor	`32 × 52 × $85 = $141,440`	modeled	$0.14M
3	Close	`6 × 8 × 85 × 12 = $48,960`	modeled	$0.05M
4	Total	`450,000 + 141,440 + 48,960 = $640,400`	modeled; record has no drift dollar	≈$0.6M/yr
5	Double-count guard	`if S1's $0.9M oversell row is on the table, drop row 1 here → $190,400`	rule	≈$0.2M

SOLUTION FLOW — THE SYSTEM & DATA MAP



SENSITIVITY — LOW / BASE / HIGH

INPUT	LOW	BASE	HIGH
Postings a day × 0.5% unmatched allowance	1,000 → 5	2,000 → 10	5,000 → 25
Cancel rate (30M × $15)	0.05% → $225K	0.10% → $450K	0.25% → $1.125M
Bridge hours/week × 52 × $85	16 → $70,720	32 → $141,440	48 → $212,160
Modeled total	≈$0.35M	≈$0.64M	≈$1.39M
Late line size on a 12-unit posting	1 → AFS −1	2 → −2	whole posting → −10 (parked)

SPOKEN STATEMENT

> "S/4 posts a store's sales as one nightly aggregate — one quantity, one cost. The stores sell all day. So I gave the sold-but-not-posted units their own bucket, Shipped-Not-Billed: up at the ring, the omni sale and the ship confirm, keyed so a replay is a no-op; down only when the aggregate posting matches — aggregation number to header to lines, parked unless the quantities sum. Three units rung at 10:02 are off the promise at 10:02, not at 23:30. The matcher pro-rates the posting's cost back to each line — cost times line quantity over posting quantity, last line absorbs the rounding — so $300 over twelve units in five lines is 75, 50, 100, 25, 50, and $100 over three lines is 33.33, 33.33, 33.34. Financial On-Hand is published once from ISM, with SNB as its own column, and ties to MB5L at month-end; cost stays S/4's. If a posting will not match, SNB stays high and we under-sell — never over-sell. Two Financial On-Hand truths became one, Finance owns the aging report, and the dollar view is about $0.6M a year, modeled."

IF CHALLENGED

> **"The cancel rate is borrowed from S1 — double-counted?"**
>
> Different window: S1's rate is the vocabulary problem across node types; this is the posting-lag window inside one node. If the room wants one number, I keep S1's and drop this row — the mechanism still stands on the record.

> **"32 hours of Finance time — source?"**
>
> Benchmark, not record. The record is the mechanism: the bridge spreadsheet died when its owner owned the real report. A two-week time study on Inventory Accounting prices it properly.

TWIST MAP — LIKELY LP QUESTIONS FROM THIS INTERVIEWER → THE ANGLE TO TAKE

LIKELY QUESTION (THIS INTERVIEWER)	LP	ANGLE — WHAT TO EMPHASIZE	SKIP / NEVER SAY
Tell me about getting two functions onto one number.	Earn Trust	Same on-hand from the physical side first; Finance owns the aging report.	Never say Finance was wrong to keep a bridge.
A time you put the customer first in a system design.	Customer Obsession	Decrement at the sale; under-sell is the only failure direction.	Never claim a measured cancel reduction.
What did you hand off, and how did you know it stuck?	Ownership	Aging report run by Inventory Accounting without me; month-end ties through ISM.	Days-to-tie is not in the record.
A metric you created.	Dive Deep	Match rate within 24 h and SNB aging p95 — design targets with owners.	Never present 99.5% as a result.

BEHAVIORAL PACK — FOUR SEPARATE 90-SECOND STAR ANSWERS, EACH FROM A DIFFERENT MOMENT OF THE STORY NEW

EARN TRUST · AIQB Q5 **Q1. "Tell me about a time when your team's goals were out of alignment with another team you relied on in order to meet your goal. How did you work with the other team?"**

My goal was one published on-hand; Finance's goal was a close they could defend, and their answer was "we keep our own reconciliation." They were not wrong to distrust Operations' timing — S/4 saw a store sale only at the nightly aggregate, and anything before that was somebody's spreadsheet. So instead of asking them to trust a new number, I showed them the same number arriving from the physical side first: the store ring adds to Shipped-Not-Billed at 10:02, S/4 Available stays 60 until the posting, and the Financial On-Hand row they read carries both columns — 60 and 12 — so they can see tonight's posting before it happens. Then the posting matches and both columns move together: 48 and 0. What made it theirs was the exception path: anything that will not match ages on a report with a 24-hour flag and a 48-hour Sev, and Inventory Accounting owns that report. Their UAT entrance was "every SIT case ties to the cent," and it did. Two truths became one because the bridge spreadsheet's owner got the real report.

Number to land: FOH truths 2 → 1; 60 / 12 → 48 / 0 · **Pivot to NACF:** When Finance and Ops disagree on inventory at an FC, I would show the physical event first and give Finance the exception report to own. · **Do NOT say:** that Finance resisted change — they had a real control need; the signing-org names (user anchor).

CUSTOMER OBSESSION · AIQB Q8 **Q2. "Tell me about a time when you had to balance the needs of the customer with the needs of the business. What did you do? What was the result?"**

The business need was an ERP that posts sales as a nightly aggregate — correct, auditable, and slow; the customer need was that a unit sold in a store at 10:02 stop being promisable online at 10:02. I did not fight the ERP; I added a bucket in front of it. Shipped-Not-Billed increments in real time at the ring, so AFS at the store goes 60 to 57 within the sixty-second publish budget, and the nightly posting later clears it without moving AFS at all — 48 before, 48 after. The balance point was the failure direction: when a posting will not match, I chose to leave SNB high and under-sell rather than guess and over-sell — the business sees a little over-ordering by Replenishment, the customer never sees a cancel. The modeled value is about $450,000 a year of phantom-availability cancels on an illustrative 30-million-order base at $15 each — and I label it modeled, because the record is the mechanism, not a cancel count.

Number to land: 60 → 57 at 10:02; 48 = 48 across the posting; $450K modeled · **Pivot to NACF:** A customer promise date is the same contract — I would decrement availability at the earliest physical signal, never at the accounting one. · **Do NOT say:** that cancels fell by a measured amount; "$2.5M" anywhere in this story.

OWNERSHIP · AIQB Q5 **Q3. "Describe a time when you had to transition a project you owned to a new owner. What steps did you take to make sure the transition went smoothly?"**

The thing I was transitioning was not the matcher — it was the exception report, and the new owner was Inventory Accounting. Before go-live I wrote what the unmatched-aging report had to prove — every S/4 aggregate posting matched to its header and lines, or aged with a cause: late line, header mistaken for lines, return without its original — at what grain, with what tolerance: 24 hours to a flag, 48 to a Sev. Their UAT entrance was that every case ties to the cent, so they had already run the arithmetic themselves. At cutover the opening SNB was seeded from the unposted ship confirms and the exit bar was "aging report empty or explained at T+1." In hypercare I read it with them every morning until they read it without me; the last step was month-end, where Financial On-Hand tied to MB5L through the ISM number rather than a bridge spreadsheet. I do not have a canonical days-to-tie figure and will not invent one; the handoff test was that nobody called me.

Number to land: > 24 h flag, > 48 h Sev; aging report empty or explained at T+1 · **Pivot to NACF:** I would hand every inventory exception report at an FC to a named owner with a tolerance before launch. · **Do NOT say:** a days-to-tie number; that I still run the report.

DIVE DEEP · AIQB Q6 **Q4. "Have you ever created a metric that helped identify a need for a change in your department? What was the metric? Why did you create it?"**

The metric was match rate — aggregate postings matched to their lines within 24 hours, divided by postings received — with SNB aging p95 beside it. I created it because the old world had no number for "how much of today's sales does S/4 not yet know about, and how much of that will never reconcile." The design target was 99.5% within 24 hours; at two thousand postings a day that is at most ten unmatched, each one on an aging report with a cause. The metric did two things. It made the failure direction visible — an unmatched posting keeps SNB high and under-states the promise, so the cost of a bad match rate is Replenishment over-ordering, not a customer cancel — and it told us where the breaks were: late POS lines, a header-level feed mistaken for lines, returns without originals. I would not present 99.5% as a result; it is the target that gave Inventory Accounting something to own.

Number to land: ≥ 99.5% within 24 h (design target); ≤ 10 unmatched/day at 2,000 postings · **Pivot to NACF:** I would want the same metric between receive confirmations and inventory postings at an FC: matched within a window, aged with a cause. · **Do NOT say:** that 99.5% was achieved; that the posting volume is record.

SAME STORY, THREE ANGLES — DATA · CONFLICT/INFLUENCE/BACKBONE · FAILURE-LEARNING NEW

DATA ANGLE · DIVE DEEP

"Twelve units, five lines, three hundred dollars — to the cent, and the drift is zero."

Headline metric: 75 + 50 + 100 + 25 + 50 = 300.00; drift 48 − 48 = 0

Grain SKU × node × sales line rolled to Aggregation#; the park rule and the last-line rule are the two pieces of logic. Honest limit: no canonical drift dollar exists.

CONFLICT / INFLUENCE ANGLE · EARN TRUST

"Finance kept its own reconciliation until the real report was theirs."

Headline metric: FOH truths 2 → 1

The influence move was showing the same on-hand from the physical side first and making the aging report Finance's, with its UAT entrance "ties to the cent." Cost ownership stayed with S/4 — a concession that bought the signature.

FAILURE / LEARNING ANGLE · ARE RIGHT, A LOT

"Under-stating is safe — and it is also invisible."

Headline metric: −2 never +2; but −10 on a parked posting is Replenishment's problem

The safe failure direction means a bad match rate shows up as over-ordering, not an alarm. Do differently: wire the aging report into Replenishment's exception view from day one, not just Finance's.

confirm whether Replenishment read the aging repo

TRAP DRILLS — THE THREE NASTIEST FOLLOW-UPS, WITH NUMERIC ANSWERS NEW

Definitional — why isn't SNB in on-hand? Or is it?

It is inside Finance's book and outside the promise. Until the posting, S/4 Available still holds the 12 sold units, so any on-hand built from Available carries them; SNB is its own column — 60 and 12 — so Finance sees what tonight's posting removes. AFS subtracts it: 60 − 12 = 48. After the match both fall together: 48 and 0. The two canon phrasings of FOH differ on whether SNB is "added"; the arithmetic that cannot be wrong is that the sold unit is counted once.

Say: *"SNB is the reconciling column between Finance's book and the promise — in the row Finance reads, out of the number dotcom reads, and never counted twice."* **Verify #2**

Definitional — what if S/4 and ISM disagree at T+1 on a store?

At a store the only sold-not-posted bucket is SNB, so a T+1 difference is a posting that did not match or a line that did not arrive. Example: S/4 posted 12, ISM matched 10 → SNB 2, ISM Available 48 = MB52 48 (Test A passes) but the aging report shows the 2-unit line at 24 hours — that is the control, not the drift check. If Test A itself breaks, it is a posting after the watermark or a duplicate — timing or defect, re-cut or replay, never an edit to the book.

Say: *"At a store, T+1 disagreement lives on the aging report, and it is a Sev at 48 hours with a cause, never a plug."*

Who owns the publish of Financial On-Hand — ISM or S/4?

ISM publishes FOH because it holds the sold-not-posted units S/4 will only see at the aggregate; S/4 owns the posted book, the settlement authority and cost. It is one truth because the two are forced to agree — SNB decrements only on a matched posting, Σ FOH ties to MB5L before Finance closes, tolerance zero in units, and any value gap is a WAC question for S/4.

Say: *"ISM publishes it; S/4 owns the book and the cost; the month-end tie is what makes it one number."*

20-SECOND VERSION + NACF BRIDGE NEW

20-second version. S/4 posts store sales as a nightly aggregate, so I gave sold-but-not-posted units their own bucket, Shipped-Not-Billed — up at the ring, down only on a matched posting — de-aggregated the posting's cost back to each line to the cent, and published one Financial On-Hand that Finance and Operations both read and that ties to MB5L. Two truths became one; the failure direction is under-sell, never over-sell; about $0.6M a year, modeled.

NACF bridge. NACF's equivalent is the gap between a physical event and its inventory posting — a receive confirmation, a transfer, a vendor return — and whether placement and Finance read one on-hand or two. I would put the same three rules in: decrement at the physical signal, clear only on a matched posting, and an aging report with a named owner and a 24/48-hour clock.

Safe-say box. "A unit sold at 10:02 is off the promise at 10:02, not at the nightly posting — and if the posting will not match we under-sell, never over-sell." · "Two Financial On-Hand truths became one; the dollar view is about $0.6M a year, modeled — the record carries no drift dollar."

SOURCE TRACE — every quantitative claim in the three stories NEW

Status key: **Y** = value found as stated · **Y-r** = same fact, rounding/derivation · **CONFLICT** = sources disagree (safe wording used) · **UNSUPPORTED-anchor** = Amazon/GFiber packet only, no Apple v11 or primary doc (user anchor) · **modeled / illustrative** = arithmetic on labeled inputs, never record. Page refs: GFiber = GFiber_MASTER_Packet_v8_f158.txt [p#]; Apple = Apple_PMIII_Interview_Stories_v11 [p#]; cb91 = amazon-l6-scm-master-packet_cb91.txt [p#]; recon = NUMBERS-RECON.md row.

NUMBER / CLAIM	STORY	SOURCE FILE	PAGE / ROW	STATUS
DEFINITIONS AND FORMULAS				
OOB = S4_Avail − WMS_OnHand; Basis = min(S4_Avail, WMS_OnHand); asymmetry (negative OOB never promised)	**S1-R, card**	GFiber v8; cb91	GFiber p38–39 "Math"; cb91 p20	**Y**
AFS = Basis − PickRes − EcomPick − EcomPack − SNB; MerchRes/LF outside S4_Avail	**all**	GFiber v8; cb91; Packet 3 U3/U6	GFiber p39; cb91 p20	**Y**
FOH_n = Avail + MerchRes + LostFound (DC); store: no MerchRes	**S1-R, S3-R, card**	GFiber v8; cb91	GFiber p39 and R11 p44; cb91 p20 l.608	**Y**
FOH "available plus merchandise reserve plus lost-and-found plus SNB" (S3 walkthrough)	**S3-R, card**	GFiber v8; cb91	GFiber p63; cb91 p21 flow box	**CONFLICT with the S1 formula on the SNB term → safe sentence (Verify #2)**
"FOH" expands to Financial On-Hand (never "free on hand")	**all**	GFiber v8; cb91; Lan packet	GFiber p39, p62–64; cb91 p20–21	**Y (user's "free of holds" phrasing: no source — Verify #1)**
Worked family 1,000 / 980 / OOB +20 / 200 / 120 / 60 / 40 / 560; flip to 1,020 → 580 not 600	**S1-R**	GFiber v8; cb91; Packet 3 U3/U6	GFiber p39 "Worked example"	**Y (illustrative in source)**
Store walk: ring 3 at 10:02 → SNB 40→43 / replay no-op / overnight posting → drift 957 − 957 = 0	**S3-R (adapted to 60 → 48)**	GFiber v8	p39	**Y (pattern; figures re-cut, illustrative)**
Drift = ISM.S4_Avail − MB52 per SKU × SLoc; tolerance 0; "Sev, not a planning variance"	**all**	GFiber v8; cb91; Packet 3 U6	GFiber p39, R2 p42; cb91 p20	**Y**
Test B identity: S4_Avail = (AFS − P) + PickRes + EcomPick + EcomPack + SNB + max(OOB,0)	**S1-R**	GFiber v8	p41 (spoken), R2 p42	**Y**
R2 rows: late GR +24 (17:03 vs 17:00 watermark; MB5B re-cut); duplicate 601 −10 with Test B still green	**S1-R W3/W4**	GFiber v8	R2 worked example p42–43	**Y**
WMS feed ≤ 15 min; late ⇒ publish suppressed, last-good held and flagged; completeness ≥ 99.9%	**S1-R, S1-R-infra**	GFiber v8	data-flow table p37; mapping p39; "what goes wrong" p44	**Y**
Event → republish p95 < 60 s; key MATNR × node × channel × version; consumers never compute	**all**	GFiber v8; cb91	GFiber p38, p40; non-functional p46	**Y**
POS real-time Kafka; late > 24 h flagged; duplicate dropped by key	**S3-R**	GFiber v8	p37, p40	**Y**
Aggregate posting nightly / intraday batch; decrement on match only; SNB never negative	**S3-R**	GFiber v8; cb91	GFiber p38, p40, p63; cb91 p21	**Y**
R8: park unless Σq_i = q_P; dup header dropped; > 24 h flag, > 48 h Sev; failure direction safe	**S3-R, S1-R W3**	GFiber v8	R8 p43–44	**Y**
COGS_i = C_P × q_i ÷ q_P; last line absorbs rounding; 5 lines / 12 / $300.00 → 75, 50, 100, 25, 50; late line → SNB 2, AFS −2	**S3-R**	GFiber v8; cb91	GFiber p63; cb91 p21	**Y**
$100.00 → 33.33 / 33.33 / 33.34; $100.00 over 3,2,2 → 42.86 / 28.57 / 28.57	**S3-R W2**	— (arithmetic from the canon rule)	—	**illustrative (recomputed)**
Match rate ≥ 99.5% within 24 h = design target; SNB_aging_p95	**S3-R**	GFiber v8; cb91; recon S3.03	GFiber p63; cb91 p21	**UNSUPPORTED-anchor → "design target," never a result**
Movement types 101/311/601/641/653/655/701/702, 344 blocked transfer; MARD/MSEG/MATDOC/MB52/MB5B/MB5L/MB5T/MBEW-VERPR	**all**	GFiber v8	p37–45 (standard SAP named there; 344 is standard SAP, not named in source)	**Y (344: standard-SAP label)**
Mapping row "Commit → Picking Reserve; pick confirm → Ecom Pick; pack → Ecom Pack" vs R4 "Σ open STO = PickRes (record); Σ open orders = EcomPick"	**card, S1-R**	GFiber v8	p39 vs R4 p43	**CONFLICT → R4 used; Verify #3**
R3 "a Pack balance at a DC must be 0 by definition" vs canon walk Ecom Pack 60 at the DC	**card**	GFiber v8	R3 p43 vs p39	**CONFLICT → walk follows canon figures; Verify #4**
Returns post to unrestricted (653/655), RTV runs as STO — excluded from the S1 record	**card**	GFiber v8	R1 p42	**Y**
R6: WMS snapshot includes allocated/picked/packed (presence), shipped-not-posted netted via SNB; same-cutoff rule	**S1-R setup, W3**	GFiber v8	R6 p43	**Y (the "980 = 940 + 40" reading is this packet's interpretation — Verify #5)**
R7: if AFS < 0 publish 0, write row; clean at T+1 = go-live exit bar	**S1-R, S1-R-infra**	GFiber v8; Packet 3 U6	R7 p43	**Y**
R10: three-column balance; in-transit as its own batch inside a freeze; no SKU published from two locations; zero-then-publish	**S1-R-infra**	GFiber v8; Packet 3 U5	R10 p44	**Y**
Control owners: Negative AFS → IC + Internal Audit + Finance; AFS Correction → ISM app; drift/OOB → IC; unmatched → Inventory Accounting	**all**	GFiber v8; Packet 3 U6	"Who owned a break?" p44	**Y**
Issues-log row 716: 400 of 1,800 Dept 96 SKUs negative in MCS, 0 in BY (22.2%)	**S1-R T3**	UDC_Consolidated_Issues_Log row ID 716 (Comments col.); Packet 3 U6		**Y (record-from-docs; UDC log, umbrella program)**
PROGRAM DATES, SCALE, OUTCOMES				
PO-creation go-live Jan 2025 → combined May 2025 ("January planning go-live, May execution")	**all**	Apple v11; recon S1.07	C1 p22; C4 p28	**Y**
E2E chain Inbound 11/22 · Dotcom 12/20 · Retail 1/30	**S1-R**	Apple v11; recon S1.06	C1 p22	**Y**
Oversell recurring → zero customer-facing incidents through both go-lives	**all**	Apple v11; resume; recon S1.03	E1 p33; C1 p21; resume p1	**Y (no baseline count — never invent one)**
FSD v1.1 3/5/25 → v1.2 5/16/25 → signed 6/5/25; three signing orgs	**S1-R, S3-R**	GFiber v8 only; recon S1.04/S1.05	GFiber l.969, 3107	**UNSUPPORTED-anchor → "first week of June" if unsure; names only if confident**
60+ stakeholders; about a dozen impacted systems	**S1-R**	Apple v11; recon S1.10, S2.10	C1 p22; C4 p28	**Y**
$1M MCS support retired; ≈$2.5M/yr headline	**S1-R, S1-R-infra**	SAP_Transformation_Interview_Playbook p2 p2 (Adobe prep); GFiber v8; recon S1.01		**UNSUPPORTED-anchor → always split anchor vs modeled**

NUMBER / CLAIM	STORY	SOURCE FILE	PAGE / ROW	STATUS
Modeled $1.53M = 30,000,000 × 0.10% × $30 ($900K) + 3 × $110K ($330K) + 40 × 60 × $125 ($300K)	**S1-R**	cb91 S1 model rows; recon S1.02/S1.12	recomputed	**modeled (recomputed, matches)**
≈4 FTE reconciling by hand → ≈1	**S1-R**	Packet 3 U6; cb91 S1	benchmark	**modeled / benchmark**
$30 per oversell; $4.75 undersell (= $19 × 25%)	**S1-R, S1-R-infra**	Packet 3 U6 / U3	—	**benchmark / illustrative (identical to Packet 3)**
FOH truths 2 → 1; over-sell from unmatched = 0 by construction	**S3-R**	GFiber v8; recon S3.01/S3.02	GFiber p62; cb91 p20	**UNSUPPORTED-anchor → "two truths to one"**
≈$0.6M/yr = $450,000 + $141,440 + $48,960 = $640,400	**S3-R**	cb91 S3 model rows; recon S3.04	cb91 p20; recomputed	**modeled (recomputed, matches)**
Cutover runbook 343 tasks / 48,885 task-min ≈ 815 h (814.75)	**S1-R-infra**	Apple v11; resume; recon S1.09/S2.12	C2 p24	**Y-r**
Rollback triggers sync < 99% × 4 h / p95 > 2× × 1 h; armed both go-lives, never fired	**S1-R-infra**	Apple v11; resume; SAP Narrative p4; recon S2.13	C2 p24; resume p1	**Y**
Cutover-night clock (20:00 → 01:00 → 06:00)	**S1-R-infra T2**	— (steps record; times not)	—	**illustrative clock on record steps**
Runbook effort 814.75 × $110 = $89,622.50 ≈ $90K	**S1-R-infra**	Packet 3 U5	—	**illustrative rate**
RF devices "hundreds"; 3–4-day estimate; script failed on refreshed hardware; device test days before training; training +1.5 weeks; date held	**S1-R-infra**	Apple v11; recon S18.09/S18.10	G2 p44–45	**Y (count: ~400 vs ~100 CONFLICT in source → "hundreds")**
What-if rule → UDC RF-device proof-of-concept before cutover week	**S1-R-infra**	Apple v11	G2 p45 ("On UDC that thinking became the RF-device proof-of-concept before cutover week (Chapter C1)")	**Y**
Mocks 7.0% → 1.8% → 0.3% vs 0.5% published before mock 1	**S1-R-infra**	Apple v11; SAP Narrative p6; recon S18.03	D2 p31	**Y**
~$1.8M, ~3% under (≈$54K); $20M pipeline; ≈$2.5M/month = $20M ÷ 8	**S1-R-infra**	Apple v11; recon S18.02/S18.04/S18.05	D1 p28–29	**Y-r ($54K and $2.5M/month derived; "not banked")**
72-hour in-transit freeze; RP read-only 30 days	**S1-R-infra**	Apple v11; recon S18.06	D1 p29, D2 p31	**Y**
Hypercare exit < 2 blockers/day, week 3 of 4	**S1-R-infra**	Apple v11; recon S18.07	D1 p29	**Y (S18 program figure, used as the pattern)**
Dock utilization = scheduled ÷ (doors × hours × rate), cap ~85%; hours = units ÷ UPH × gross-up	**S1-R-infra**	amazon-nacf-inbound-booklet (CAP-B, CAP-D)	capacity section	**Y (framework; inputs 10 × 16 × 1.0, 130, 120 UPH, 1.2 illustrative)**
12 EA/h sell rate; 3 EA per 15 min; 12 EA per hour; 1,000 events/h; 240-unit receipt; 700-unit damage; −30	**S1-R, S1-R-infra**	—	—	**illustrative (recomputed)**
Master S4 failure beat: in-transit rule reversed; σ ≈ 0.3 day → ~0.49-day window (1.645 × 0.3)	**S1-R angle card**	GFiber v8; recon S4 / canon #56	recon row 56	**UNSUPPORTED-anchor (math correct)**
ROUTER AND BEHAVIORAL PACKS				
AIQB wordings: Dive Deep Q1/Q6; Invent and Simplify Q1; Have Backbone Q2; Earn Trust Q1/Q5; Insist on the Highest Standards Q1; Are Right, A Lot Q4; Bias for Action Q4; Deliver Results Q2; Customer Obsession Q8; Ownership Q5	**all**	Consolidated_questions-AIQB-Reference-Guide-GLO-EN_0418.txt	LP sections, numbered lists	**Y (verbatim)**

VERIFY BEFORE YOU SAY IT — terms that stayed ambiguous, each with the safe sentence NEW

Last-resort list: items absent from every source in /tmp/sources and the master, or where two canon passages disagree. Confirm with your own memory before the loop; if unsure, say the safe sentence.

#	ITEM	WHERE USED	STATUS	SAFE SENTENCE IF UNSURE
1	**"FOH" expansion. Every source says *Financial On-Hand*. The brief's phrase "on-hand free of holds" appears nowhere in /tmp/sources or the master.**	Cram sheet, card, S1-R, S3-R	absent from sources	"In my program FOH meant Financial On-Hand — Finance's number: Available plus merchandise reserve plus lost-and-found. If your FOH means free-on-hand — unrestricted stock free of holds — that is what we called Available, S/4's unrestricted book."
2	**Does FOH include SNB? S1 formula: FOH = Avail + MerchRes + LostFound. S3 walkthrough: "… plus lost-and-found plus SNB." Until the posting, S/4 Available still contains sold units, so adding SNB would count them twice unless "available" there means the posted book less SNB.**	Card, S1-R W2 row 8, S3-R trap 1	CONFLICT inside canon	"Financial On-Hand is Available plus merchandise reserve plus lost-and-found, with Shipped-Not-Billed carried as its own reconciling column so the sold unit is counted once and Finance sees what tonight's posting removes; it ties to MB5L at month-end."
3	**Where a web-order commit lands: GFiber mapping row says "Commit → Picking Reserve"; R4 (record) says Σ open STO = PickRes and Σ open orders = EcomPick. This packet follows R4 (web commit → EcomPick; STO → PickRes).**	Card, S1-R W1 E3/E4	CONFLICT inside canon	"A store transfer enters Picking Reserve when the STO is created; a web order enters the ecom buckets when it is committed — and the tie-out is Σ open STO = Picking Reserve, Σ open orders = Ecom Pick, tolerance zero."
4	**Ecom Pack at a DC: R3 says "a Pack balance at a DC must be 0 by definition"; the canon walk carries Ecom Pack 60 at the DC. This packet keeps the canon figure and treats Pack as the packed-not-ship-confirmed window.**	Card, S1-R W1	CONFLICT inside canon	"Ecom Pack is the packed-not-yet-shipped window; at a DC it clears on ship confirm, so its end-of-day balance should be zero."
5	**The reading "WMS 980 = 940 physically pickable + 40 shipped-not-posted" (R6 "shipped-not-posted netted via SNB") is this packet's interpretation to keep the ship-confirm step consistent; the canon example does not spell it out.**	S1-R setup, E6–E7, trap 1 of S1-R-infra	interpretation	"For the out-of-balance comparison the floor count is netted for units that have shipped but not yet posted, so a ship confirm never manufactures a phantom OOB."
6	**FSD dates 3/5/25 → 5/16/25 → 6/5/25 and the three signing-org names (Supply Chain Strategy, Inventory Management, FSC Inventory Accounting) — GFiber v8 only.**	S1-R, S3-R phases	UNSUPPORTED-anchor	"Signed in the first week of June 2025; three orgs signed."
7	**$1M MCS support retired — Adobe-prep playbook and GFiber only.**	S1-R results	UNSUPPORTED-anchor	"About a million a year of legacy support cost retired — my program number."
8	**Cutover-night clock times (20:00 → 01:00 → 06:00) and the 1,000 events/hour trigger volume.**	S1-R-infra Table 2	illustrative	"The order of steps is record; I would not quote the hours."
9	**Whether Replenishment read the unmatched-aging report (failure-learning angle in S3-R).**	S3-R angle card	not in record	"What I would do differently is wire the aging report into Replenishment's exception view — I don't have a record that we did."
10	**Whether a cause class was recorded on every AFS Correction audit row (carried from Packet 3 U6).**	S1-R trap 2	not in record	"Every correction had a screen, a role and an audit row; the cause class is what I would make mandatory on the row."
11	**Days-to-tie at month-end (FOH → MB5L).**	S3-R	not in record	"The mechanism is canonical; I don't have a days-to-tie figure and won't invent one."
12	**Physical-to-S/4 posting latency for an RF scan in seconds (not in any source; only the ≤15-min WMS feed and <60-s republish budgets are).**	S1-R-infra	absent from sources	"The budgets I owned were the fifteen-minute floor feed and the sixty-second republish; the scan-to-WMS hop was the WMS team's and I did not set a number on it."

Never say (this add-on). "free on hand" as the program's term · a bare "$2.5M" or "$1.5M" without its noun · "100 of 400 devices" or any cost multiplier · "the ISM issues log" (the 836-row log is UDC's) · "99.5% match rate" as a result · "fifteen tracks" · "20+ systems" · any BOM / component framing · the unit counts of the walk as program data.